Investigating the role of Nonsense Mediated Decay in Embryonic Stem cell differentiation

PDN Part II Experimental Project

Alessa Weiler

Department of Physiology, Development and Neuroscience

University of Cambridge

This dissertation is submitted for the degree of

Bachelor of Arts

Trinity College Academic year 2022-2023

Declaration

I confirm that the material in this dissertation is not copied from any published material, unless it is clearly identified as such and a full source reference given. I confirm that the material in this dissertation is not a paraphrase or abstract of any published material, unless it is clearly identified as such and a full source reference given. I understand that the examiners may use any means available to test for plagiarism within my dissertation. I confirm that, other than where indicated, this dissertation is my own work. I confirm that the word count is 8253.



Alessa Weiler

Academic year 2022-2023

Acknowledgements

I would like to express my thanks to my supervisor, Dr. Claire Senner, for her exceptional mentorship and unwavering support. Her belief in my potential has been a driving force in my academic journey, pushing me to improve my abilities and fostering my growth as a scientist and as an individual. Dr. Senner has not only taught me the fundamentals of scientific research design and implementation and the importance of critical thinking but has also instilled in me both curiosity and a profound appreciation for the intricacies of the world of molecular and developmental biology. Her commitment to excellence and her ability to balance both empathy and rigour has set a high standard for me to emulate in my future endeavours.

Abstract

Embryonic development is a complex process that requires intricate regulation of gene expression. Nonsense-mediated decay (NMD) is a cellular mechanism that degrades faulty RNA molecules to prevent the production of non-functional proteins. The *Smg1* gene plays a key role in the NMD cascade, and its absence leads to embryonic lethality. This study aimed to elucidate the role of *Smg1* in embryonic development and placental lineage specification, as well as to identify potential downstream genes affected by *Smg1*.

By studying *Smg1-/-* embryos, we confirmed the embryonic lethality reported by previous studies and further characterized the placental phenotype. We observed the absence of spongiotrophoblast and chorion layers in *Smg1-/-* placentas, accompanied by an increased abundance of trophoblast giant cells (TGCs). These findings suggest a potential role for *Smg1* in placental lineage decisions and embryonic development.

We also conducted RNAseq analysis of embryonic stem (ES) and trophoblast stem (TS) cells under stem cell conditions and differentiating conditions to investigate the differential expression of NMD-influenced genes. Results indicated that all NMD genes, as well as genes identified to be influenced by NMD showed high expression in both stem cell conditions and in differentiating conditions. Furthermore, the genes that were identified to be differentially expressed in stem cell conditions compared to in differentiating conditions in ES cells were different from those identified in TS cells, suggesting that genes downstream of *Smg1* may

play distinct roles in different cell populations and lineages. These results were confirmed by analysis of transcriptomic data which showed that different cell populations may have distinct mechanisms of dependence on NMD. However, further investigation is needed to strengthen these conclusions, including the analysis of more placentas and the examination of earlier time points in embryonic development.

Overall, our study contributes to the understanding of *Smg1*'s involvement in embryonic development and highlights the need for further investigation to elucidate its molecular mechanisms.

Table of contents

List of figures **xi**

Nomenclature **xiii**

1 Introduction **1**

1.1 Nonsense mediated decay 1

1.1.1 NMD recognizes its targets through a combination of actions from the ribosome, the Exon-Junction Complex (EJC) and the 3'UTR . . 1

1.1.2 NMD is important in development 2

1.2 SMG1 is an essential part of the NMD cascade 4

1.3 Using stem cells to study development 5

1.4 Objectives 7

2 Observations/Results **9**

2.1 In vivo analysis of *Smg*$1^{-/-}$ mouse embryos and placentas at E9.5 9

2.2 Bioinformatic analysis 12

2.2.1 RNAseq analysis to assess the expression of core NMD factors and factors downstream of NMD in ES and TS cells 12

2.2.2 TS cell analysis 18

2.2.3 Transcriptome analysis of the expression of key NMD factors shows that all factors are highly expressed, however different mechanisms of dependence on NMD may occur in different lineages 21

3 Discussion **23**

3.1 In vivo results 23

3.1.1 NMD is essential in embryonic development 23

3.1.2 NMD is crucial in the placental trophoblast lineage 25

3.2 Bioinformatic analysis 28

3.2.1 RNAseq analysis 28

3.2.2 Endoderm explorer 31

3.3 General Discussion 33

4 Materials and Methods **35**

4.1 Embedding and sectioning mouse placentas 35

4.2 H & E staining 36

4.3 Transcriptome analysis 36

4.4 RNAseq datasets analysis 36

5 Conclusion **39**

References **41**

Appendix A [Genes investigated] **47**

Appendix B Differentially Expressed Genes **79**

List of figures

1.1 NMD models 3

1.2 Deriving Stem Cells From The Early Embryo 6

2.1 Phenotypic characterization of *Smg*$1^{-/-}$ mouse embryos at E9.5 10

2.2 Phenotypic characterization of Smg1-/- mouse placentas at E9.5 11

2.3 Gene lists curated 13

2.4 Genes that are candidates to be influenced by *Smg*$1^{-/-}$ and subsequent breakdown of the NMD pathway. 17

2.5 Genes that are candidates to be influenced by *Smg1* knockout and subsequent breakdown of the NMD pathway. 20

2.6 Endoderm Explorer 22

Nomenclature

Roman Symbols

AS	Alternative Splicing
DEG	Differentially Expressed Gene
E	Embryonic day
EJC	Exon-Junction Complex
ESC	Embryonic Stem cells
ExE	Postimplantation extraembryonic ectoderm
FISH	Fluorescence In Situ Hybridization
ICM	Inner Cell Mass
MEF	Mouse Embryonic Fibroblast
NMD	Nonsense Mediated Decay
PrE	Primitive Endoderm
PTC	Premature Termination Codon

SynT Syncytiotrophoblast

TGC Trophoblast Giant Cell

TS Trophoblast Stem Cells

Chapter 1

Introduction

1.1 Nonsense mediated decay

Nonsense Mediated Decay (NMD) is a mechanism that occurs after the transcription of DNA into messenger RNA (mRNA) to regulate gene expression [18]. Canonically, the NMD pathway's role is to detect mRNA strands that harbour premature termination codons (PTCs) - a nucleotide triplet that terminates the translation of mRNA prematurely - and degrade these transcripts. More recent research has demonstrated NMD's implication in the maintenance and regulation of the stability of physiological mRNAs including protein-coding RNAs, long-noncoding RNAs and microRNAs [14, 43], and in regulating alternative splicing events (AS) that can lead to the introduction of PTCs, referred to as AS-coupled NMD [51, 37]

1.1.1 NMD recognizes its targets through a combination of actions from the ribosome, the Exon-Junction Complex (EJC) and the 3'UTR

According to the ribosome release model [6], ribosomes translating normal mRNA release the polypeptide chain at the stop codon, however when a ribosome encounters a PTC, it

becomes stalled and triggers degradation of the aberrant mRNA by recruiting NMD factors. Recruitment of NMD factors occurs through SMG1-mediated phosphorylation of UPF1, which leads to repression of further translation, triggering mRNA degradation by recruitment of nucleases either directly (eg by SMG6 for endonucleolytic decay) or indirectly (eg through the SMG5-SMG7 heterodimer in exonucleolytic decay) (1.1) [40]. The rate of ribosome release can be influenced in physiological transcripts by elements such as the length of the 3'UTR and the presence of cis-acting elements. The efficiency of translation termination determines whether a ribosome releases or stalls. Further research is necessary to better understand the role of NMD in development and the transcripts which it targets.

1.1.2 NMD is important in development

NMD has been shown to be implicated in cell differentiation, embryonic development and tissue morphogenesis. By modifying the quantity of specific mRNA transcripts, NMD enables the appropriate timing and expression of genes required for these processes [38, 53]. NMD's implication in development and embryogenesis is all the more evident when observing various NMD knockout mice and stem cell lines. Null mutations in NMD-related genes— $Rent1^{-/-}$/$Upf1^{-/-}$, $Upf2^{-/-}$, $Smg1^{-/-}$, $Smg5^{-/-}$, $Smg7^{-/-}$ and $Smg6^{-/-}$— leads to early postimplantation embryonic lethality in mice [31, 38, 50, 39, 5]. These findings suggest that NMD is essential for early mammalian embryonic development.

This is equally true in stem cell models. When hESCs were cultured under conditions promoting their differentiation into definitive endoderm, the expression of the majority of NMD-factor genes decreased [35]. On the other hand, when differentiation was directed towards ectoderm and mesoderm, there was a significant upregulation of various NMD-factor

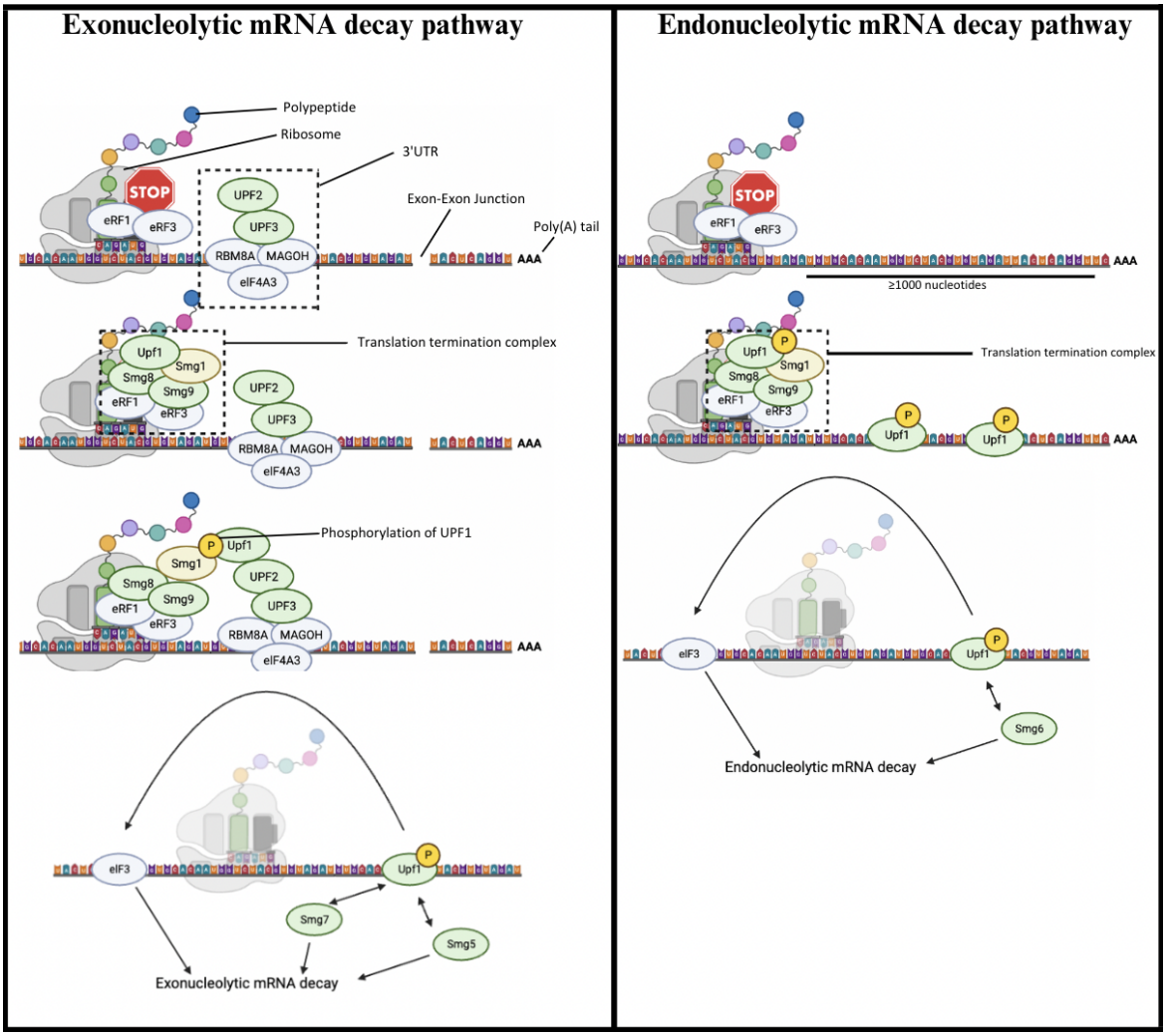


Fig. 1.1 Exonucleolytic and endonucleolytic NMD cascade. Pre-mRNA splicing occurs in parallel to the deposition of an EJC and EJC-associated NMD factors such as UPF3X. In the case of EJC-mediated NMD, translation termination at a PTC that is located further than 50-55 nucleotides upstream of an exon-exon junction leads to the EJC not being dislocated by the ribosome. The SMG1-UPF1-eRF1-eRF3 (SURF) complex is then formed by UPF1, SMG1, SMG8 and SMG9 joining with the translation termination complex. Phosphorylation of UPF1 by SMG1 triggers exonucleolytic mRNA degradation by recruitment of nucleases indirectly by the SMG5 and SMG7 heterodimer. The translation termination complex and UPF1 still play an important role in the detection and degradation of aberrant mRNA in the endonucleolytic mRNA decay cascade, however these factors harness the 3' untranslated region (UTR) as a signal to initiate NMD [26]. UPF1 phosphorylation of SMG1 leads to the recruitment of SMG6, leading to endonucleolytic mRNA decay

genes [35]. Similarly, a wildtype mouse ESC line showed different gene regulation patterns when it was guided to develop into the three primary germ layers [35].

1.2 SMG1 is an essential part of the NMD cascade

SMG1 is part of the phosphoinositide kinase-like kinase (PIKK) family of proteins and is one of the key elements of the nonsense mediated decay (NMD) pathway [10]. It is required for the phosphorylation of UPF1, a key component of the NMD pathway, which then leads to subsequent degradation of the PTC-containing mRNA [22, 55]. Knockdown of *Smg1* in human cells has been shown to lead to the upregulation of NMD factors UPF1 and UPF2 – pointing to its role in the regulation of downstream factors in the NMD pathway [56]. The importance of SMG1 in RNA quality control and gene control pathways is true for mosses such as Physcomitrella patens [33] through to mice [38] and humans [56], and has been shown to be conserved across the eukaryotic domain [34].

Dissociation and expansion of head, limb and visceral tissues of E13.5 $Smg1^{-/-}$ mice to form mouse embryonic fibroblasts (MEFs) have demonstrated that the NMD pathway in these MEFs is not intact [38]. Full knockouts of *Smg1* in mouse models are undeniably embryonic lethal between E8 and E13 [38]. Other data confirm that $Smg1^{-/-}$ embryos are embryonic lethal by E8.5 [46, 15, 28]. Though current evidence suggests an early post-implantation lethality, it is unclear at exactly what time point *Smg1* deficiency influences development, as well as how *Smg1* influences development in the context of its role in the NMD pathway. Most recently, the Deciphering the Mechanisms of Developmental Disorders (DMDD) study [45] analyzed 103 mouse knockout lines with embryonic lethality or sub-viability and uncovered a higher prevalence of placental phenotypes than previously recognized. One of these placental phenotypes observed was that of the $Smg1^{-/-}$ mice, which was reported to be embryonic

lethal by E14.5, and sub-viable by E9.5. In this study, the aim was to further examine the placental phenotype associated with the Smg1 knockout, and explore the root causes of the embryonic lethality phenotype.

1.3 Using stem cells to study development

Given the early stage of development in which $Smg1^{-/-}$ embryos arrest, one should consider the use of in vitro models for further analysis. Molecular analysis is difficult to conduct when using small amounts of material, and is all the more difficult as the embryo is younger and smaller in size. Furthermore, a large number of published data from cell lines is available to analyze. For these reasons, it is compelling to use stem cells from early embryonic development to study the different embryonic lineages.

Before implantation, totipotent cells from the zygote differentiate into the inner cell mass (ICM) and the trophectoderm (TE). The ICM further differentiates into the epiblast, which gives rise to the embryo proper and the primitive endoderm (PE), which gives rise to the yolk sac, whereas the TE eventually forms a large proportion of the placenta 1.2. Embryonic stem (ES) cells are derived from the ICM of the early embryo [36]. These cells are pluripotent (can differentiate into any cell type), and can self-renew indefinitely [48]. Isolation and culture of ICM cells in serum (BMP4) and leukemia inhibitory factor (LIF) leads to the establishment of an ES cell line [42]. ES cells are a suitable model to recapitulate in vivo stem cells of the embryonic lineage [23]. Trophoblast stem (TS) cells are derived from the TE of the early (E3.5-4.5) embryo, the extraembryonic ectoderm (ExE) of the E6.5 embryo or the chorion of the E7.5-E8.5 embryo. Isolation and culture of these cells in heparin, FGF4 and MEF-conditioned TS medium leads to the establishment of a TS cell line. Mouse TS cells can differentiate into any cell subtype of the trophoblast lineage (multipotent) [47], and

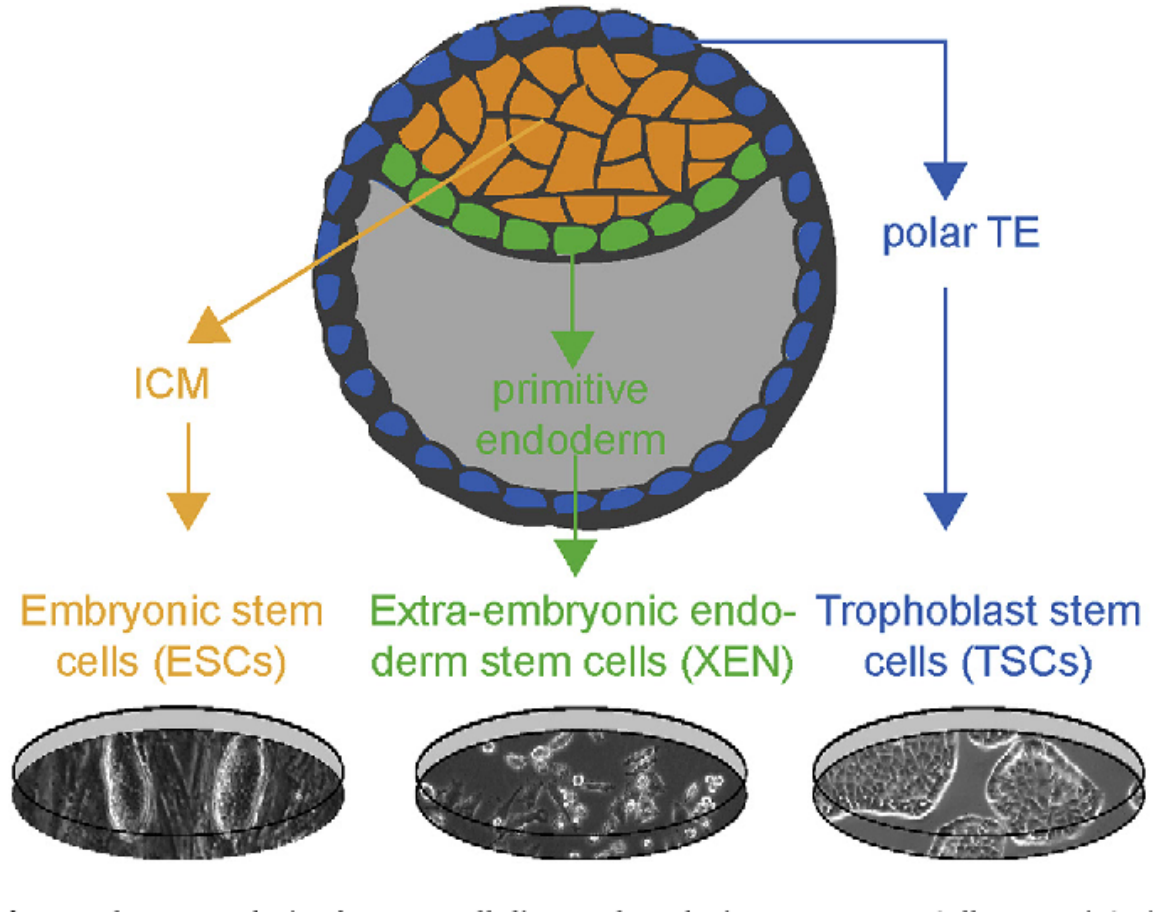


Fig. 1.2 Overview of the three types of stem cells derived from the early embryo and their respective origins. The figure illustrates the location of XEN cells, ES cells, and TS cells during early embryonic development. XEN cells (Xenogeneic Endoderm Stem Cells) originate from the primitive endoderm, ES cells (Embryonic Stem Cells) are derived from the inner cell mass, and TS cells (Trophectoderm Stem Cells) are derived from the trophectoderm. Figure from Kubacza et al., 2017 [24]

are a suitable in vitro model to recapitulate the in vivo stem cells in the placenta, in the ExE and the chorion [25].

To confirm the embryonic and placental phenotypes in the DMDD study, we dissected pregnant mice at E9.5 and sectioned the placentas of these *Smg*1$^{+/-}$ intercross litters given that no embryo was present. In these placentas, we observed an overgrowth of trophoblast giant cells (TGC) and no perceivable spongiotrophoblast layer or chorion. Further bioinformatic analysis of ES and TS cell datasets comparing their gene expression in stem cell and

differentiating conditions revealed that all NMD factors were highly expressed, however, a list of differentially expressed genes identified to be influenced by NMD were elucidated. Analysis of single-cell transcriptome data revealed that different embryonic lineages may use different mechanisms of dependence on NMD.

1.4 Objectives

1. To confirm the embryonic lethality phenotype of *Smg*$1^{-/-}$ embryos and to characterize the placental phenotype from litters of intercrosses of *Smg*$1^{-/-}$ mice

2. To curate a list of genes downstream of *Smg1* that may be contributing to the embryonic lethal phenotype

3. To observe the expression levels of NMD factors in the early embryonic lineages (E3.5-4.5)

Chapter 2

Observations/Results

2.1 In vivo analysis of *Smg*$1^{-/-}$ mouse embryos and placentas at E9.5

Smg$1^{-/-}$ mice have previously been described as exhibiting embryonic lethality but there is no agreement about the timing, with one study describing arrest shortly after implantation and another reporting recovery of embryos as late as E12.5. As the DMDD project described a placental phenotype at E9.5 [45], it was decided to begin analysis at this time point. Litters from intercrosses of *Smg*$1^{+/-}$ mice were imaged by Dr. Senner at E9.5, revealing a severe phenotype in *Smg*$1^{-/-}$ conceptuses. No embryonic material was recoverable within the small yolk sac observed (2.1). This data indicates that *Smg1* is necessary for early murine embryogenesis prior to E9.5 and supports earlier postimplantation lethality.

To further analyze the placental phenotype, a portion of the decidua where the implantation site is located was removed. These implantation sites were fixed, embedded, and sectioned and H&E staining was carried out to observe the structures present. This histologi-

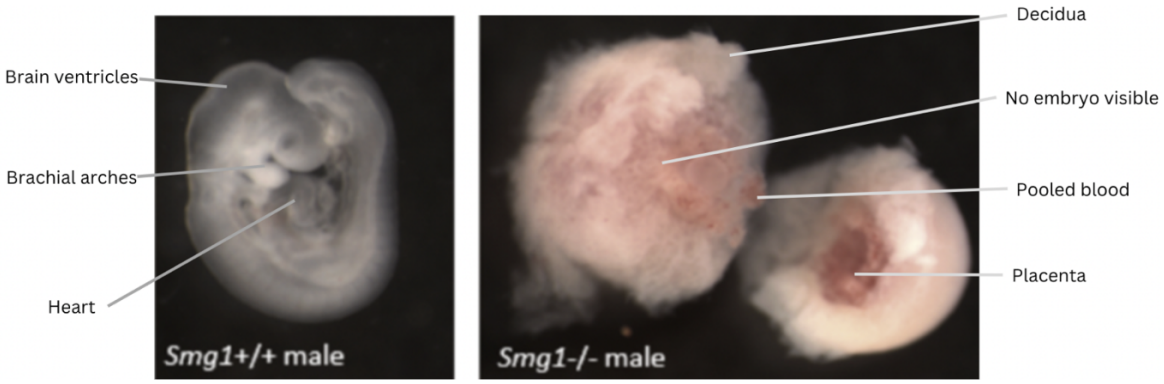


Fig. 2.1 Embryos from intercrosses of *Smg*1$^{+/-}$ mice were imaged at E9.5 by Dr. Senner, revealing a more severe phenotype than previously reported, with the embryo essentially absent alongside *Smg*1$^{-/-}$ placentas. This observation suggests that *Smg1* is critical for early murine embryogenesis prior to E9.5.

cal analysis demonstrated that male and female *Smg*1$^{+/+}$ and *Smg*1$^{+/-}$ placentas developed normally but *Smg*1$^{-/-}$ placentas showed severe defects, in which only one cell population was present.

Initiation of branching morphogenesis through the chorion and the beginning of labyrinth formation was clearly visible in wildtype and heterozygous placentas, suggesting that fetal blood vessels would be formed in the labyrinth at a later time point. This was not the case in homozygous placentas, which lacked a chorion altogether, although pooling of blood can be seen in the decidua and in the group of trophoblast giant cells (TGCs). This is in line with observations by McIlwaine et al. [38] that there was no development of the vascular system, however, hematopoiesis does initiate.

The ectoplacental cone, which forms the spongiotrophoblast layer and a layer of TGCs, was clearly visible in *Smg*1$^{+/+}$ and *Smg*1$^{+/-}$ placentas. In *Smg*1$^{-/-}$ placentas, the ectoplacental cone and the spongiotrophoblast layers were not present, and a surplus of TGCs formed a sphere-like shape as opposed to the laminar style in the controls. This data suggests that the differentiation of placental cells is influenced by *Smg1*, with *Smg1* enabling correct

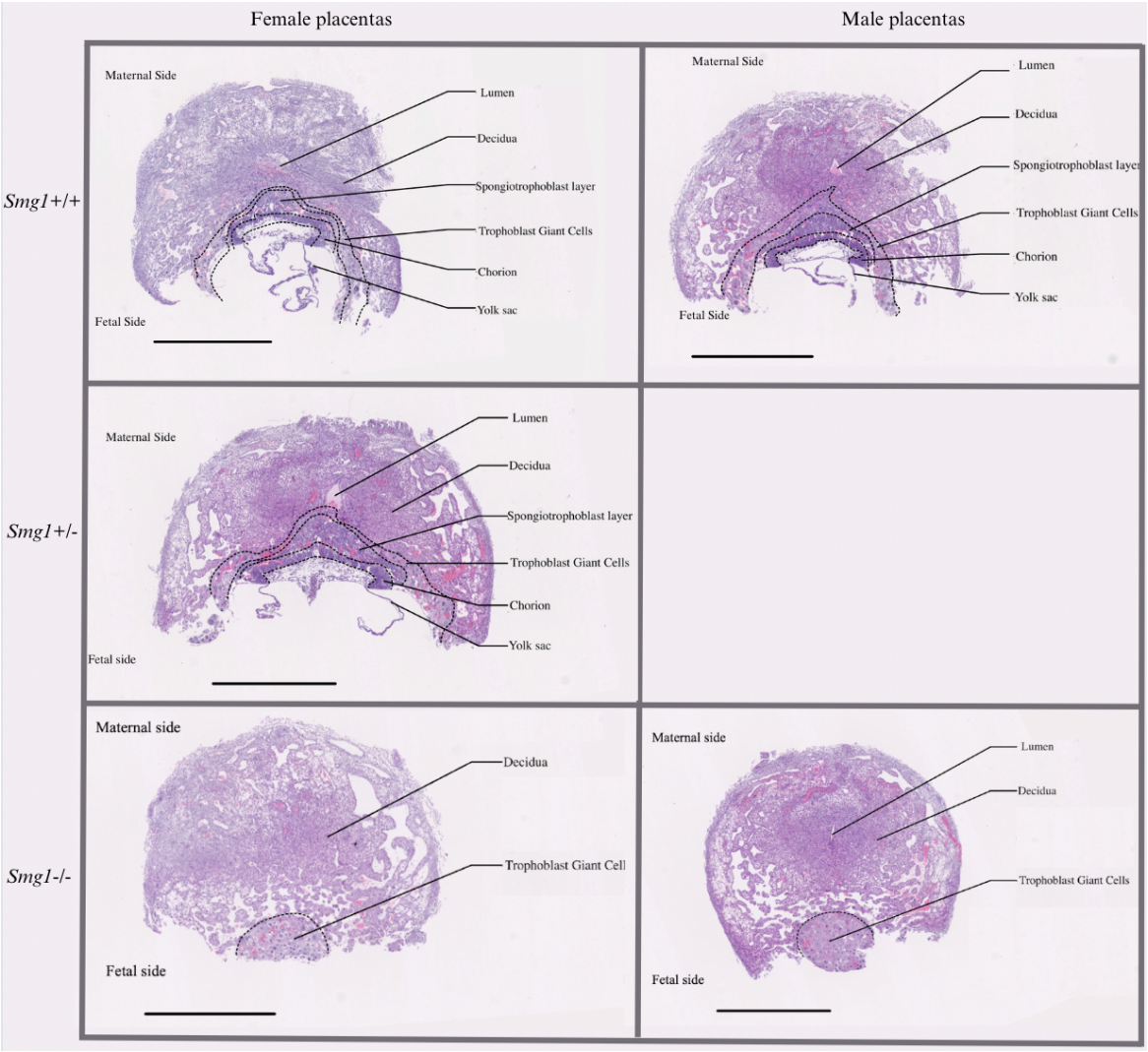


Fig. 2.2 Placentas from intercrosses of *Smg*1$^{+/-}$ mice were sectioned and stained, revealing a more severe phenotype than previously reported, with a marked absence of the chorion and spongiotrophoblast cells, and an abundance of trophoblast giant cells in homozygous placentas.

proliferation and differentiation of all trophoblast cell lineages. Further molecular analysis combined with research focused on the derivation and differentiation of *Smg*1$^{-/-}$ TS cells is ongoing in the lab to further understand this differentiation defect.

2.2 Bioinformatic analysis

As described above, it appears that embryonic lethality in *Smg*1$^{-/-}$ conceptuses is occurring in the early postimplantation period. In order to understand how *Smg1* and the NMD pathway may be influencing differentiation of the embryonic and trophoblast compartments, a bioinformatic analysis of previously generated data was carried out.

2.2.1 RNAseq analysis to assess the expression of core NMD factors and factors downstream of NMD in ES and TS cells

To gain a better understanding of the underlying molecular mechanisms and identify potential pathways involved in embryonic and placental defects, RNA sequencing data from ES and TS cells were utilized. These stem cell models are particularly valuable for investigating differentiation events, as they offer a controlled environment to closely examine the molecular changes that occur during early developmental processes.

To identify genes downstream of NMD that were also differentially expressed during differentiation relative to stem cell conditions, four lists of potentially important genes were analyzed and criteria were set to produce distinct lists (2.3)

The four lists are specified in Appendix 1, but can be summarised as follows:

1. Key components of the NMD cascade, as identified through a literature review

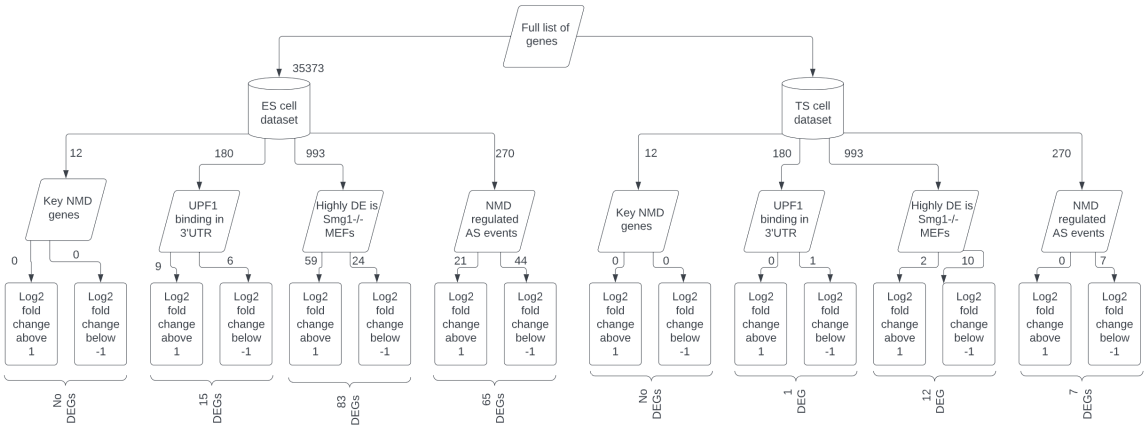


Fig. 2.3 Flow diagram displaying the gene lists assessed, and the resulting differentially expressed genes identified from these lists. Numbers indicate the genes that belong to the list below the arrow.

2. UPF1 target genes that bind to the 3'UTR, as identified in supplemental table S5 from Hurt et al. [16]

3. Genes known to be AS in Smg1 KO MEFS (supplementary information from McIlwain et al. [38])

4. Genes known to be differentially expressed in Smg1 KO MEFs (supplementary information from McIlwain et al. [38])

For each gene list, two lists were produced. The first list included genes with higher expression in stem cells, referred to as "TS>TS DIFF" in the TS cell dataset or "ES>ES DIFF" in the ES cell dataset, and may be critical in maintaining stem cell state or defining cell identity. The second list included genes with higher expression in differentiating cells, referred to as "TS DIFF>TS" in the TS cell dataset or "ES DIFF>ES" in the ES cell dataset. These may be crucial in trophoblast/ICM differentiation or may have a role in placental/embryonic development and function.

RNA sequencing data analysis involves alignment of the data to a reference genome to identify the genes that are expressed in the sample, quantification of the expression of each

gene by counting the amount of reads that map to each gene and normalizing these counts to compensate for dissonances in sequencing depth, differential expression analysis to identify genes that are differentially expressed between different conditions, and functional analysis to gain insights into the biological processes that are influenced by modulations in gene expression (gene ontology analysis, pathway analysis or other forms of enrichment analysis). Given the dissonance in the number of differentially expressed genes, with as little as one gene being identified in one dataset, we chose not to perform this final step, however all other steps were performed in a systematic manner using the Seqmonk tool.

ES cell analysis

Key NMD genes All key NMD genes studied were highly expressed in stem cell conditions and in differentiating conditions. This was expected because NMD is an essential process for maintaining cellular homeostasis and ensuring mRNA quality control. Furthermore, none of the genes investigated displayed significant levels of differential expression in stem cell conditions relative to differentiating conditions for 24 hours. Overall, these findings suggest that NMD is highly active and important in both stem cell and in differentiating conditions (2.4a).

NMD-mediated alternative splicing events predicted to introduce a PTC Genes that belong to this list may activate the NMD pathway through the introduction of a PTC during alternative splicing. Like the key NMD genes, genes from this list had high levels of expression in stem cell conditions and in differentiating conditions. Sixty-five differentially expressed genes were identified from this list. Of these differentially expressed genes, twenty-two were upregulated during differentiation. Notable differentially expressed genes include

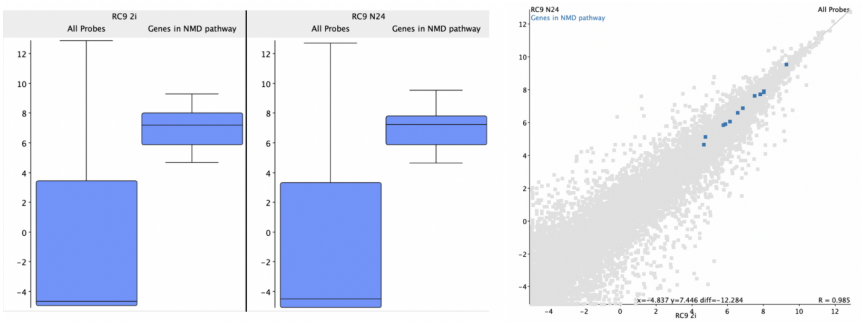
Snrpn - a gene implicated in neural development, autism and schizophrenia [30, 41], and previously shown to be modulated by NMD [13] (2.4b)

UPF1 binding in the 3'UTR This gene list investigated encapsulated genes in which UPF1 can bind in the 3'UTR of mRNA, suggesting targeting by NMD. High levels of expression in differentiating and in stem cell conditions can be seen in this list. Nine out of the fifteen differentially expressed genes were more highly expressed in differentiating conditions than in stem cell conditions. Notably, *Esrrb* – a gene with an important role in maintaining the pluripotent state [11] - was more highly expressed in differentiating conditions. Interestingly, this gene was upregulated in E5 $Smg6^{-/-}$ embryos, suggesting that its expression may be modulated by the NMD pathway [31] 2.4c

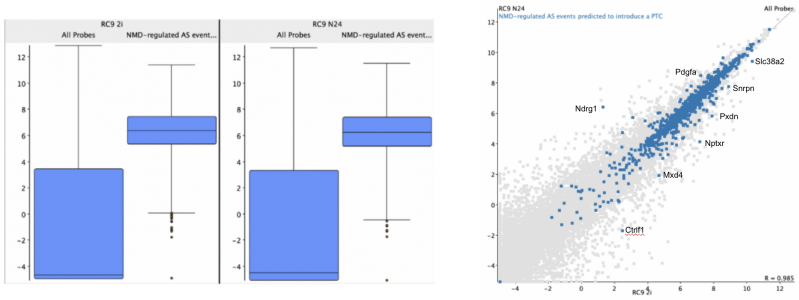
Highly differentially expressed genes in $Smg1^{-/-}$ Mouse Embryonic Fibroblasts (MEFs) Genes from this list were found to be potentially influenced by *Smg1*, as the absence of *Smg1* leads to a change in their expression in MEFs [38]. The mean expression levels of genes from this list were higher than the mean expression levels of all probes, however this dissonance was not as striking as in other lists investigated. 59 of the 83 differentially expressed genes identified were upregulated in differentiating conditions. Notably, *Otx2* – a transcription factor involved in neural development [3] - and *Bmp7* – a protein involved in the formation of bones and cartilage [2] – were identified to be differentially expressed in this dataset. 2.4d

Overall, these findings suggest that genes suspected to be influenced by *Smg1* are highly expressed in stem cells and in differentiating conditions. Genes from these lists that were identified to be differentially expressed in stem or differentiating conditions could potentially influence ES cell differentiation and may play a role in the $Smg1^{-/-}$ embryonic lethal phenotype. These differentially expressed genes warrant further investigation to better

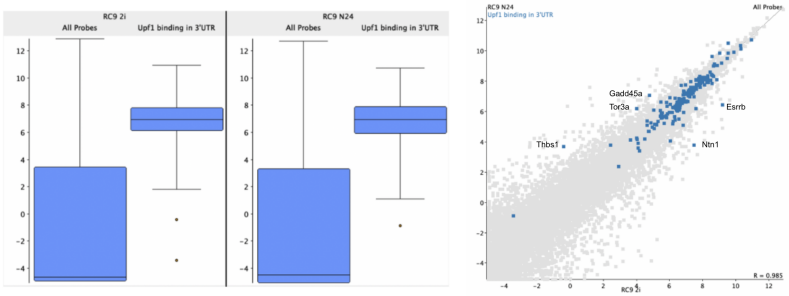
understand the molecular cascade that mediates embryonic lethality, and to understand molecular cascades that interact with *Smg1*.



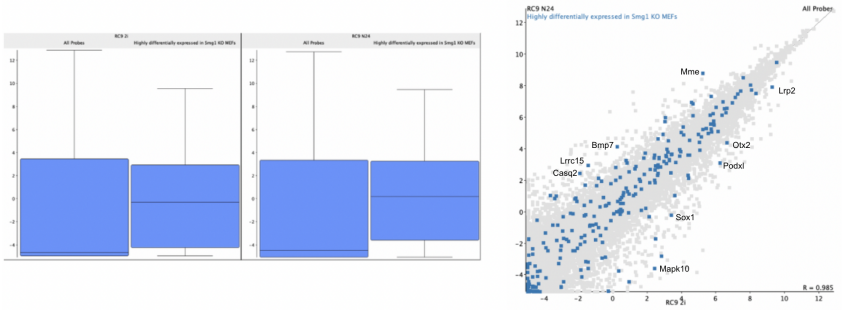
(a) Differential expression of key genes involved in the NMD pathway in ES cells in stem cell and in differentiating conditions



(b) Differential expression of NMD-mediated alternative splicing events predicted to introduce a PTC in ES cells in stem cell and in differentiating conditions



(c) Differential expression of genes in which UPF1 can bind in the 3'UTR in ES cells in stem cell and in differentiating conditions



(d) Differential expression of genes highly differentially expressed in *Smg*$1^{-/-}$ MEFs in ES cells in stem cell and in differentiating conditions

Fig. 2.4 Genes that are candidates to be influenced by *Smg*$1^{-/-}$ and subsequent breakdown of the NMD pathway.

2.2.2 TS cell analysis

Key NMD genes NMD factors are highly expressed in both differentiation and stem cell conditions. None of the key NMD genes were significantly differentially expressed in stem cell conditions or in differentiating conditions, suggesting that NMD is important in both conditions in trophoblast cells. 2.5a

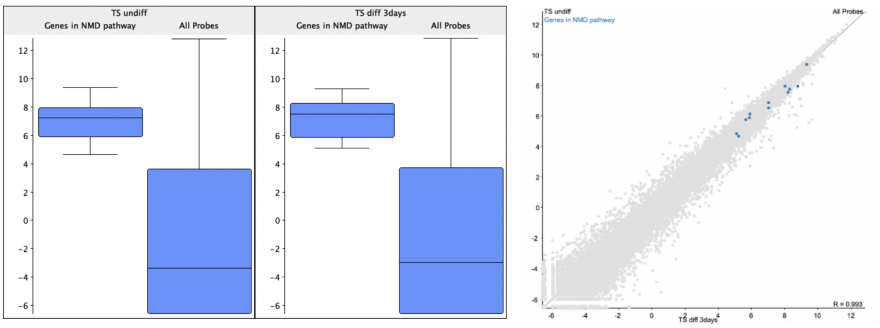
NMD-mediated alternative splicing events predicted to introduce a PTC Seven differentially expressed genes were identified at the $p<0.05$ level of significance, of which all genes were downregulated in differentiating conditions relative to stem cell conditions. Furthermore, a comparison of this differentially expressed gene list with the differentially expressed genes identified in ES cells showed that there is no overlap between these lists. Notably, *Eif4a2* – a gene linked to a neurodevelopmental disorder and important in neural tube formation [52] - was identified as differentially expressed in this dataset. 2.5c

UPF1 binding in the 3'UTR High levels of expression were observed in differentiating and in stem cell conditions. Only one differentially expressed gene was identified, *Rassf1* – a tumour suppressor gene [9]. This differentially expressed gene was downregulated in differentiating conditions relative to stem cell conditions and was not identified as differentially expressed in the ES cell dataset. 2.5d

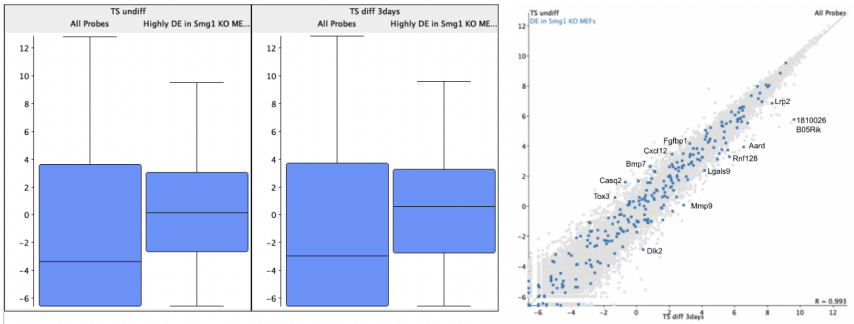
Highly differentially expressed genes in $Smg1^{-/-}$ Mouse Embryonic Fibroblasts (MEFs) Genes from this list had expression levels similar to those of all probes. Unlike ES cells, in which 70% of all differentially expressed genes were found to be upregulated during differentiation, only approximately 16% of all differentially expressed genes in this gene list were found to be upregulated during differentiation in the TS cell dataset. This was the only

gene list to show an overlap of differentially expressed genes identified in both ES and TS cells: *Bmp7*, *Casq2*, *Slc1a3*, and *2610307P16Rik*. 2.5b

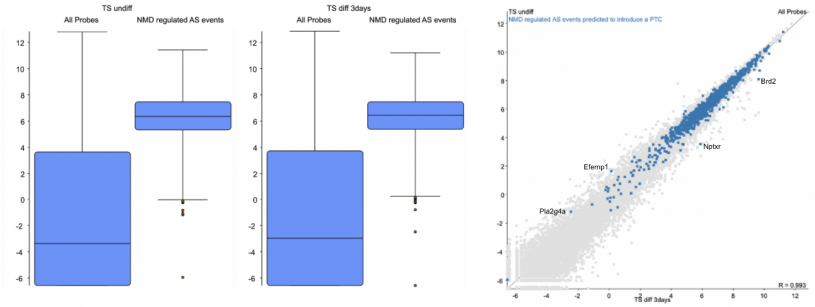
Overall, these results show that genes that may interact with Smg1 and are differentially expressed in stem cell conditions relative to differentiating conditions in TS cells tend to be upregulated in stem cell conditions. This suggests that the gene may contribute to the maintenance of the stem cell state, or may contribute to the characteristics of stem cells such as self-renewal and pluripotency.



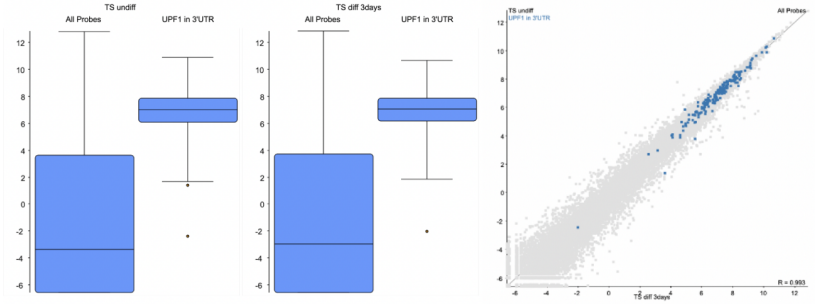
(a) Differential expression of key genes involved in the NMD pathway in stem cell and in differentiating conditions



(b) Differential expression of genes highly differentially expressed in $Smg1^{-/-}$ MEFs in stem cell and in differentiating conditions



(c) Differential expression of NMD mediated alternative splicing events predicted to introduce a PTC in stem cell and in differentiating conditions



(d) Differential expression of genes in which UPF1 can bind in the 3'UTR in stem cell and in differentiating conditions

Fig. 2.5 Genes that are candidates to be influenced by *Smg1* knockout and subsequent breakdown of the NMD pathway.

2.2.3 Transcriptome analysis of the expression of key NMD factors shows that all factors are highly expressed, however different mechanisms of dependence on NMD may occur in different lineages

Previously obtained data from over 110 single-cell transcriptomes from all lineages of E3.5-E4.5 mouse embryos was analyzed and visualized. The data was originally collected to map the differentiation trajectories of the mammalian endoderm in the mouse embryo, generated by Nowotschin et al. [44]. It was produced by isolating cells from sequentially staged mouse embryos between E3.5 and E8.75 for single-cell RNA sequencing. We analyzed the single-cell transcriptome data representing all early embryonic lineages within the mouse embryo from E3.5-E4.5. Expression patterns of each key gene in the nonsense mediated decay pathway were visualized in the early E3.5 embryo (see Appendix 1 for the list of NMD genes investigated) as force-directed layouts. Three controls were also included in the analysis - *Nanog* (high expression in the ICM), *Cdx2* (high expression in the TE) and *Gata6* (high expression in the primitive endoderm (PrE)) (Figure 2.6).

Comparing the expression levels of key NMD genes with these control genes using force-directed graphs, we observed that overall, all NMD genes were highly expressed throughout all lineages (2.6). However, some of these genes were more highly expressed in one lineage compared to another. SMG5 and SMG7 form a heterodimer and indirectly recruit nucleases to initiate mRNA decay as part of the endonucleolytic NMD cascade, whereas SMG6 is a key component of the exonucleolytic NMD pathway. *Smg5* and *Smg7* are most highly expressed in the TE (mean imputed log gene expression in the TE for *Smg5* = 4.36 at E3.5, 4.1 at E4.5; mean imputed log gene expression in the TE for *Smg7* = 3.86 at E3.5, 3.98 at E4.5), and were

most lowly expressed in the PrE. On the other hand, SMG6 was most lowly expressed in the TE (mean imputed log gene expression in the TE = 2.15 at E3.5, 1.9 at E4.5). This suggests that although all NMD genes are highly expressed throughout the embryonic lineages, there may be preferential mechanisms of RNA decay in different embryonic lineages.

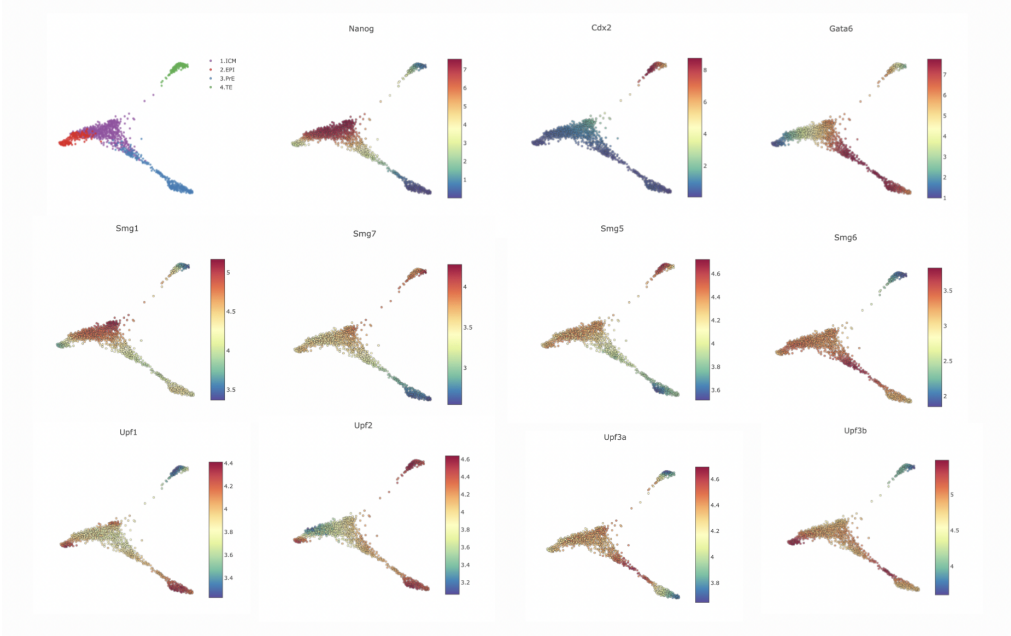


Fig. 2.6 All key NMD factors identified were highly expressed throughout the early embryonic (EE) lineages. Note that the scale bars- representative of imputed log gene expression- for each graph are different, as the factors involved in NMD are highly expressed throughout the early embryo. Data from Nowotschin et al.[44] was analyzed using their computational approach to visualize the transcriptional profiles of different genes in the NMD pathway in all EE lineages of the early (E3.5-4.5) embryo. The resulting force-directed graphs were then arranged in three rows. Not all force-directed graphs are depicted here. **a.** Controls. **b.** Expression levels of *Smg5, Smg6* and *Smg7* **c.** Representative examples of the high expression of NMD factors

Chapter 3

Discussion

Placental dissections of littermates from an intercross of *Smg*$1^{+/-}$ mice were performed to elucidate the role of *Smg1* in placental lineage specification. Bioinformatic analysis of RNAseq datasets from ES and TS cells in stem cell conditions and in differentiating conditions, as well as analysis of single-cell transcriptome data from the early embryo, was performed to identify potential downstream genes that may contribute to the *Smg*$1^{+/-}$ embryonic and placental phenotypes. Our research provides novel insights into the role of *Smg1* in embryonic development and lineage specification. We identified the potential involvement of SMG1 in lineage differentiation throughout the embryo, as well as *Smg1*'s distinct roles in different cell populations and lineages.

3.1 In vivo results

3.1.1 NMD is essential in embryonic development

Our embryo and placental dissection results reveal that the *Smg*$1^{+/-}$ embryonic lethality phenotype occurs before E9.5. This finding is consistent with other Roberts et al's [46]

characterization of *Smg*$1^{+/-}$ embryos, as well as with the timing of embryonic lethality in other NMD mutants such as $Upf2^{-/-}$ which is embryonic lethal before E9.5.

The embryonic lethality we observed at E9.5 contrasts with McIlwain et al's observations of *Smg*$1^{+/-}$ E12.5 conceptuses. The stark contrast in the severity of our phenotype relative to theirs might be due to the mouse model of *Smg*$1^{+/-}$ used. The phenotype of the *Smg1* tm1b(EUCOMM)Hmgu mouse line that we studied replicates the phenotype observed by the Deciphering the Mechanisms of Developmental Disorders Project (DMDD) and was created using a comprehensive pipeline to guarantee complete knockout. The International Mouse Phenotyping Consortium (IMPC) provided a reliable and well-established pipeline to generate a complete knockout of the *Smg1* gene, ensuring the robustness and validity of our experimental model. The data presented here confirm the previously reported placental phenotype in the DMDD study using the *Smg1* tm1b(EUCOMM)Hmgu mouse line. The successful replication of the phenotype strengthens the rationale for using this model in our investigation of the molecular mechanisms underpinning implantation and placentation processes.

Despite this adequate mouse model, we were limited in our ability to thoroughly examine the *Smg*$1^{+/-}$ phenotype, as we only analyzed embryos from two litters. This was mainly due to the health issues encountered by our mice in the animal facility, which hindered our ability to obtain a larger sample size. Future studies with larger sample sizes and healthier animal models will be essential in confirming and expanding our understanding of *Smg*$1^{+/-}$ embryonic lethality.

To further understand the molecular mechanisms behind *Smg*$1^{+/-}$ embryonic lethality, dissections and molecular analyses (such as bulk RNA-seq or single-cell sequencing) should be conducted at earlier time points, such as E6.5 and E7.5. Additionally, genotyping of

blastocysts could be performed to confirm the presence of embryos in Mendelian ratios before implantation. These future studies will provide a more comprehensive understanding of the role of *Smg1* in embryonic development and the impact of its complete knockout.

3.1.2 NMD is crucial in the placental trophoblast lineage

Critique of our results

Given that no embryo was present at E9.5, we sectioned placentas to describe the *Smg*1$^{-/-}$ placental phenotype. Upon sectioning and H & E staining placentas from a litter of an intercross of *Smg*1$^{-/-}$ mice, we observed a severe placental phenotype, in which no chorion or spongiotrophoblast (SpT) cells could be seen. These observations suggest that *Smg1* may play a role in lineage decisions within the placenta, favouring the differentiation of SpT and chorion cells whilst suppressing the differentiation of TGCs. This observation aligns with previous research demonstrating the involvement of other NMD factors in cell differentiation [50]

In addition to the absence of SpT cells and the chorion, we also observed an abundance of TGCs in the shape of a dome rather than the typical laminar shape seen in wild-type placentas. This altered morphology may be indicative of compensatory mechanisms to maintain placental function despite the absence of SpT cells and chorion. This is similar to the effect seen in other loss-of-function models. For instance, Jeyarajah et al. [19] have shown that the loss of function of *Gcm1* which led to the absence of the SpT layer leads to an increase in TGC cells present. More research is necessary to understand the molecular mechanisms underpinning the placental cell populations and morphology of *Smg*1$^{-/-}$ placentas, and how *Smg1* influences lineage decisions in TS cells. It would be interesting to look at the role of *Smg1* in regulating genes involved in placental development through RNAseq of *Smg*1$^{-/-}$

TS cells to understand if *Smg1* interacts with other regulatory pathways that control lineage specification in the placenta.

We also observed that heterozygous placentas were comparable to wild-type placentas, and only homozygous placentas displayed a severe phenotype. This suggests that *Smg1* expression is not dose-dependent but rather functions in an all-or-nothing manner, meaning that the presence of *Smg1* on one chromosome is sufficient for normal placental function. This is in line with findings from zebrafish in which depletion of SMG1 did not affect embryonic development [53]. This finding implies that *Smg1* may act through a dominant mechanism in regulating placental development, suggesting that *Smg1* may have a threshold of expression required to perform its role in the regulation of placental development, and once this threshold is reached, any additional *Smg1* expression may not contribute to further improvements in function. This dominant mechanism of *Smg1* action may be due to the fact that the presence of *Smg1* on one chromosome is sufficient to produce the required amount of SMG1 protein to form functional complexes with other proteins involved in the NMD pathway such as UPF1 [54], ensuring proper NMD function and regulation of placental development. Further research is needed to elucidate the precise molecular mechanisms underlying the dominant nature of *Smg1* function in placental development, including the investigation of *Smg1* interactions with other NMD factors, as well as its potential involvement in other cellular processes that may be critical for placental development and function. Overall these results indicated that some cells within the trophoblast lineage were less affected than others by $Smg1^{-/-}$, in line with our bioinformatic analysis that showed different differentially expressed genes in different lineages.

Critique of our methodology

We were limited in the number of placentas analyzed due to time constraints and intermittent breeding problems in the mouse colony. As a result, we sectioned placentas from only two litters of *Smg*$1^{-/+}$ intercrosses and some placentas were damaged in processing, yielding a total of nineteen placentas for analysis. It is important to note that, for phenotyping experiments, the International Mouse Phenotyping Consortium (IMPC) recommends analyzing 28 embryos per developmental stage [1]. Therefore, our sample size was smaller than the recommended number, which should be taken into consideration when interpreting our findings. Further investigation into the role of *Smg1* in embryonic and placental development should take these guidelines into account and focus on larger sample sizes.

Research into the role of NMD in the placenta is novel and has not been previously conducted, and our findings represent novel insights into the development of extraembryonic structures of *Smg*$1^{-/-}$ mice. Given this novel approach, our primary goal was to obtain a broad overview of the organization of the mutant placentas. We opted for H&E staining due to its cost-effectiveness, simplicity, and its ability to provide this general overview of tissue morphology. Our work serves as a foundation for further analysis in this area, and future research should consider incorporating more rigorous staining techniques, such as in situ hybridization or immunofluorescence, to gain deeper insights into the molecular and cellular aspects of these structures, and to confirm the cell types present.

Future directions

To further explore the role of *Smg1* in placental lineage decisions, it would be interesting to examine *Smg*$1^{-/-}$ TS cells and to conduct a molecular analysis using techniques such as ATAC-seq and research into the methylation profile of these cells relative to a heterozygous

control [7]. Furthermore, given the observed differences in development between various trophoblast lineages, it would be interesting to investigate the effectiveness of chimeras in determining whether the absence of *Smg1* in one lineage is sufficient and in which lineage it is most important. Chimeric models can help elucidate the functional significance of *Smg1* in lineage-specific development and the potential compensatory mechanisms at play. The use of in vitro models, however, comes with its own set of limitations, as they cannot encapsulate maternal effects, which may interact with placental and embryonic development. To control for whether maternal levels of *Smg1* interact with placental and embryonic development we can perform transplantation experiments to see if there are phenotypic differences between *Smg*$1^{-/-}$ and *Smg*$1^{-/+}$ embryos in a heterozygous mother or a wildtype mother.

3.2 Bioinformatic analysis

3.2.1 RNAseq analysis

To identify genes that interact with NMD and are important in stem cell differentiation or maintenance of the stem cell state, we compared the expression levels of genes that interact with NMD in stem cell conditions with their levels in differentiating conditions. We analyzed the expression levels of these genes in RNA sequencing datasets from ES and TS cells. Future comparison of ES and TS cell gene expression in stem cell conditions would be interesting to understand the different gene pathways activated, however, this would be difficult to do in differentiating conditions because stem cell state markers are distinct in both of these lineages.

Evaluation of methodology

Our statistical analysis of the RNAseq data included using an R-based filter in Seqmonk (DESeq2) where the genes that were differentially expressed in stem cells as opposed to in differentiating conditions at the $p<0.05$ level after Benjamimi and Hochberg correction to decrease false positive rates [4]. This filtered list was then filtered further to identify genes that had a Log2 fold change between the two values above 1 or below -1. DESeq is limited insofar as it may produce false positive or false negative results if it detects outliers, or if there is a large variance in the data. This statistical test may also not encapsulate functions of genes that are co-expressed with the list investigated, or differential expression that may occur because of alternative splicing. It has been argued that a solution to these issues is to replace DESeq analysis with the Wilcoxon rank sum test [32], and it would be interesting to see if one can see similar results using this statistical test on our data instead. Furthermore, studying our differentially expressed gene list using a different data collection method such as RT-qPCR or Digital Gene Expression Profiling in an $Smg^{-/-}$ ES cell or TS cell line in stem cell conditions and differentiating conditions could help validate our results by showing that expression of these genes are dysregulated in the absence of a key NMD factor, and their expression levels differ in stem cell and differentiating conditions.

Evaluation of results

Given the difference in the conception of the ES cell and TS cell datasets, our results indicate that there was barely any overlap in the gene lists of differentially expressed genes in ES or TS cells. Most of the differentially expressed genes that were simultaneously identified in ES cells and TS cells were observed in the gene list of differentially expressed genes in $Smg1^{-/-}$ MEFs. However, this gene list might not be the most informative for understanding the role

of *Smg1* in regulating NMD throughout the embryo, because genes that are dependent on *Smg1* in MEFs may not necessarily be regulated by NMD in other cell types or developmental stages. The observed changes in gene expression due to the absence of *Smg1* in MEFs could be specific to this in vitro model and not indicative of a general mechanism across the entire embryo. Future research would benefit from examining additional datasets from other cell types and developmental stages. We were unable to find an RNA sequencing dataset for primitive endoderm cells, and this would be especially interesting to investigate given our transcriptome analysis findings (detailed below) which indicate that the primitive endoderm might use different mechanisms of dependence on NMD than the TE. It would be interesting to also look at RNA sequencing data from XEN cells to further expand our understanding of NMD and stem cell differentiation. Despite these limitations, our study sheds light on the differential expression of NMD-influenced genes in different lineages and suggests that NMD may play different roles in different cell populations.

We observed that there is a difference in gene expression patterns between embryonic stem (ES) cells and trophoblast stem (TS) cells during differentiation and in the stem cell state. In ES cells, the majority of differentially expressed genes exhibited upregulation during differentiation. This suggests that these genes might be crucial for the development and function of differentiated cell types and tissues. In TS cells, we found that most differentially expressed genes were upregulated in stem cell conditions rather than during differentiation. This observation indicates that these genes may play a critical role in maintaining the stem cell state, including characteristics such as self-renewal and pluripotency. The different gene expression patterns between ES and TS cells are suggestive of distinct molecular mechanisms governing the differentiation processes in these two stem cell types. It is essential to consider these differences when designing targeted therapeutic strategies or

studying the developmental biology of these cells. Further research is required to better understand the specific regulatory networks involved in NMD's role in maintaining stem cell identity and directing differentiation in ES and TS cells.

McIlwain et al. [38] examined gene expression in *Smg*$1^{-/-}$ mouse ES cells using RNA sequencing, conducting a differential expression analysis to compare wildtype and *Smg*$1^{-/-}$ ES cells. We delved deeper into the *Smg1* knockout's influence on stem cell maintenance and pluripotency by looking at the differential expression of candidate genes posited to be influenced by *Smg1* differentiating and in stem cell conditions, in both a wildtype ES cell and a wildtype TS cell dataset. Our results are in line with those reported by McIlwaine et al. [38], suggesting high expression of key NMD genes in stem cells and differentiating conditions. However, our work goes further by exploring key differences in differentiating and in stem cell states in a TS cell dataset as well as in an ES cell dataset, highlighting that most differentially expressed genes identified tend to be upregulated in stem cell conditions in TS cells, whereas they tend to be upregulated in differentiating conditions in ES cells. Future research should ideally combine both of our approaches, looking at gene expression in different in vitro models (ES, TS and XEN cells) of *Smg1* deficiency and comparing these with wildtype gene expression levels.

3.2.2 Endoderm explorer

Analysis of NMD factor expression using single-cell transcriptome data from the E3.5-E4.5 mouse embryo revealed that specific NMD factors have different expression levels across different early embryonic lineages. For instance, *Smg5* and *Smg7*, which form a heterodimer as part of the exonucleolytic NMD cascade, exhibit high expression in the TE and low levels of expression in the PrE, while *Smg6* - a component of the endonucleolytic cascade - is

expressed at low levels in the TE and high levels in the PrE. This analysis confirms our previous finding (detailed above) that there may be different mechanisms of dependence on one NMD pathway over another in different early embryonic lineages.

Huth et al. [17] revealed a distinct order of severity in the disruption of target gene regulation and differentiation among *Smg5*, *Smg6* and *Smg7* knockout mouse ES cells, with *Smg*5$^{-/-}$ being the most severe, followed by *Smg*6$^{-/-}$, and *Smg*7$^{-/-}$ ES cells being the least severe. Their results suggest that SMG5 and SMG7 have functions independent of their heterodimer, which contrasts with our findings of differential expression patterns of *Smg5*, *Smg7*, and *Smg6* in the early embryo suggesting that different NMD pathways may be preferentially employed in specific early embryonic lineages. This dissonance may be due to the different approaches to answer a similar question of what the role of these NMD factors is in early embryonic development. Huth et al. [17] only focused on the role of ES cells, whereas we took a whole-embryo approach by investigating the transcriptome data of all lineages of E3.5-4.5 embryos. Furthermore, Huth et al. [17] investigated knockout in vitro models, whereas we looked at wild-type in vivo models. The advantage of looking at in vivo models over in vitro models is that we have a lower risk of artifacts, and compensatory mechanisms may be at play in knockout models. Our results provide a baseline for comparison, as very little research exists which investigates the expression of specific NMD factors in early embryonic lineages.

It is important to note that the original dataset was generated to examine gene expression in the mouse endoderm, not specifically to investigate individual NMD factor expression. To draw more definitive conclusions regarding the distinct mechanisms of dependence, it would be beneficial to create a dataset specifically designed to understand the expression patterns of these NMD factors in the early embryo. For example, performing fluorescence in situ

hybridization (FISH) could enable the observation of the expression of each NMD factor in different early embryonic lineages, however, given the high levels of NMD factors observed here, a more nuanced methodology may be more appropriate. In situ, sequencing may be more appropriate as it provides spatial information about gene expression. Furthermore, lineage tracing using genetic labelling or live imaging of fluorescent reporters can be informative to track the expression of these key NMD factors in specific cell populations over the course of developmental time.

3.3 General Discussion

Smg1 has other functions outside of NMD, and has been posited to be important in the DNA damage response [8] as it is required for the phosphorylation of many proteins in this pathway including p53 and CHK2; and in the cellular stress responses such as resistance to oxidative stress and apoptosis [46]. In addition to these roles, *Smg1* is important in regulating growth control mechanisms and injury response [12]. Our work aims to shed light on potential mechanisms that underpin the embryonic lethal phenotype and NMD dysregulation of $Smg1^{-/-}$ embryos to highlight the role of the NMD pathway in early embryonic development. We focus on the role of *Smg1* in early embryonic development in the context of its role in the NMD cascade, but further research should also be conducted in the context of the other pathways of which *Smg1* is a part.

The potential of inhibiting SMG1 with wortmannin and caffeine to rescue extracellular matrix function in vivo has been discussed in the context of genetic diseases [49]. Expanding on this approach, it may be interesting to investigate the molecular profile of cells after temporary inhibition of SMG1 in vitro cell cultures, allowing for the assessment of any subsequent changes in cellular differentiation. By comparing the differentiation processes of

treated and untreated cells, this technique could unveil novel insights into the regulatory role of SMG1 in development.

Other NMD factors including *Upf3B* are implicated in neural development anomalies including autism spectrum disorders (ASD), attention deficit hyperactivity disorder (ADHD) and schizophrenia [21]; and *Upf2* inhibition in vivo leads to behavioural and neurophysiological changes mediated by the immune system [20]. *Smg1* – and NMD more broadly – has the potential to be implicated in a range of different diseases, especially given the normal-appearing phenotype that we have observed in *Smg*$1^{+/-}$ mice. Given this potential implication, therapeutic solutions harnessing the NMD pathway can be created. It would be interesting to investigate the role of these factors in neural development specifically, given that mutations in *Upf3A*, *Upf3B*, *Eif4a3*, *Smg6* and *Upf2* are all linked to a range of cognitive impairments and neurodevelopmental disorders [29, 41].

Chapter 4

Materials and Methods

4.1 Embedding and sectioning mouse placentas

Implantation sites at E9.5 were fixed overnight at 4°C in 4% paraformaldehyde (PFA), washed three times with phosphate-buffered saline (PBS) for 10 minutes per wash at room temperature, and dehydrated in 70% ethanol overnight at 4°C. The tissue was then transferred to glass scintillation vials and dehydrated through serial ethanol solutions (70%, 100%, 100%), followed by two 90-minute paraffin washes in histoclear. The tissue was transferred to cassettes and washed two times for 90 minutes each in paraffin at 65°C. One placenta per block was embedded by placing it on its side in a metal mould, with the white cassette placed on top. The blocks were left to sit for 15-20 minutes on a cold plate before being removed from the metal moulds. The paraffin-embedded placenta blocks were then sectioned at 7µm slices in a close-to-serial manner until the midline was reached. Sections were placed in a warm water bath (42°C) to remove wrinkles and two sections were mounted per Superfrost/Polysine/X-tra slide. The slides were left to dry overnight.

4.2 H & E staining

Slides with placental sections were immersed in two washes of xylene for five minutes each, followed by a series of ethanol washes in decreasing concentration (100%, 100%, 70%, 50%, 30%) for 2 minutes each. The slides were then placed in tap water for five minutes, followed by hematoxylin staining for 3 minutes. They were quickly washed in tap water in a container and placed in acid alcohol (1% HCl in 70% Ethanol) for 10 seconds. The slides were thoroughly washed in tap water in the sink for 6-10 minutes, placed in eosin for 30 seconds and thoroughly washed in tap water for 6-10 minutes. The slides were dehydrated in a series of ethanol washes in increasing concentration (30%, 50%, 70%, 100%, 100%), for a minute per wash, followed by two five-minute xylene washes. Coverslips were mounted on the slides with DPX media and slides were left to dry in the fume hood.

4.3 Transcriptome analysis

Gene expression patterns using single-cell data at E3.5 were analyzed using Endoderm Explorer [44], an online tool which combines a range of computational methods including Harmony (correcting for batch effects), Palantir (model continuities in cell fate choices) and MAGIC (observation of gene expression changes).

4.4 RNAseq datasets analysis

Previously obtained RNA sequencing data from the GFP TS cell line ([47], Senner and Hemberger, unpublished data) in stem cell conditions (n=3) or in differentiating conditions for 3 days (n=3) was analyzed. Published RNA sequencing data from a wildtype mouse ES

cell line in stem cell conditions (2i, n=4) and in differentiating conditions (N24, n=4) [27] were also analyzed.

Data visualization, quantification of transcripts and generation of scatter plots were carried out using Seqmonk software (https://www.bioinformatics.babraham.ac.uk/projects/seqmonk/). Differentially expressed genes were identified using the DESeq function embedded in Seqmonk using default parameters and further filtered for genes with a log2 fold change >1.

Chapter 5

Conclusion

In conclusion, this research provides new insights into the role of *Smg1* in embryonic development and placental lineage specification. The embryonic lethality observed in *Smg*$1^{-/-}$ embryos was confirmed and the placental phenotype was characterised, revealing a potential role for *Smg1* in placental lineage decisions. Furthermore, RNAseq analysis of embryonic stem (ES) and trophoblast stem (TS) cells suggests that *Smg1* and other NMD factors may play distinct roles in different cell populations and lineages.

Despite these advances, several limitations and challenges remain, including the need for more extensive analysis of placentas, examination of earlier time points in development, and further investigation into the role of NMD in hormonal regulation disorders during development. Future research should also explore the use of in vitro models, chimeric studies, and machine learning algorithms to overcome the limitations of current experimental approaches.

Overall, this dissertation highlights the complex and multifaceted role of *Smg1* in embryonic development and emphasises the need for continued research to fully elucidate the underlying molecular mechanisms. Understanding the role of Smg1 and other NMD factors

in lineage decisions and embryonic development has broad implications for our knowledge of stem cell biology, developmental disorders, and regenerative medicine.

References

[1] Adams, D., Baldock, R., Bhattacharya, S., Copp, A. J., Dickinson, M., Greene, N. D. E., Henkelman, M., Justice, M., Mohun, T., Murray, S. A., Pauws, E., Raess, M., Rossant, J., Weaver, T., and West, D. (2013). Bloomsbury report on mouse embryo phenotyping: recommendations from the impc workshop on embryonic lethal screening. *Disease Models Mechanisms*.

[2] Aluganti Narasimhulu, C. and Singla, D. K. (2020). The role of bone morphogenetic protein 7 (bmp-7) in inflammation in heart diseases. *Cells*, 9(2):280.

[3] Beby, F. and Lamonerie, T. (2013). The homeobox gene otx2 in development and disease. *Experimental Eye Research*, 111:9–16.

[4] Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1):289–300.

[5] Boehm, V., Kueckelmann, S., Gerbracht, J. V., Kallabis, S., Britto-Borges, T., Altmüller, J., Krüger, M., Dieterich, C., and Gehring, N. H. (2021). Smg5-smg7 authorize nonsense-mediated mrna decay by enabling smg6 endonucleolytic activity. *Nature Communications*, 12(1).

[6] Brogna, S. and Wen, J. (2009). Nonsense-mediated mrna decay (nmd) mechanisms. *Nature Structural amp; Molecular Biology*, 16(2):107–113.

[7] Buenrostro, J. D., Wu, B., Chang, H. Y., and Greenleaf, W. J. (2015). Atac-seq: A method for assaying chromatin accessibility genome-wide. *Current Protocols in Molecular Biology*, 109(1).

[8] Chen, J.-S., Beckley, J. R., McDonald, N. A., Ren, L., Mangione, M., Jang, S. J., Elmore, Z. C., Rachfall, N., Feoktistova, A., Jones, C. M., and et al. (2015). Identification of new players in cell division, dna damage response, and morphogenesis through construction of schizosaccharomyces pombe deletion strains. *G3 Genes|Genomes|Genetics*, 5(3):361–370.

[9] Dammann, R., Li, C., Yoon, J.-H., Chin, P. L., Bates, S., and Pfeifer, G. P. (2000). Epigenetic inactivation of a ras association domain family protein from the lung tumour suppressor locus 3p21.3. *Nature Genetics*, 25(3):315–319.

[10] Denning, G., Jamieson, L., Maquat, L. E., Thompson, E. A., and Fields, A. P. (2001). Cloning of a novel phosphatidylinositol kinase-related kinase. *Journal of Biological Chemistry*, 276(25):22709–22714.

[11] Festuccia, N., Dubois, A., Vandormael-Pournin, S., Gallego Tejeda, E., Mouren, A., Bessonnard, S., Mueller, F., Proux, C., Cohen-Tannoudji, M., Navarro, P., and et al. (2016). Mitotic binding of esrrb marks key regulatory regions of the pluripotency network. *Nature Cell Biology*, 18(11):1139–1148.

[12] González-Estévez, C., Felix, D. A., Smith, M. D., Paps, J., Morley, S. J., James, V., Sharp, T. V., and Aboobaker, A. A. (2012). Smg-1 and mtorc1 act antagonistically to regulate response to injury and growth in planarians. *PLoS Genetics*, 8(3):e1002619.

[13] Gray, T. A., Saitoh, S., and Nicholls, R. D. (1999). An imprinted, mammalian bicistronic transcript encodes two independent proteins. *Proceedings of the National Academy of Sciences*, 96(10):5616–5621.

[14] Han, X., Wei, Y., Wang, H., Wang, F., Ju, Z., and Li, T. (2018). Nonsense-mediated mRNA decay: a 'nonsense' pathway makes sense in stem cell biology. *Nucleic Acids Research*, 46(3):1038–1051.

[15] Ho, U., Luff, J., James, A., Lee, C. S., Quek, H., Lai, H.-C., Apte, S. H., Lim, Y. C., Lavin, M. F., and Roberts, T. L. (2019). *SMG1* heterozygosity exacerbates haematopoietic cancer development in *Atm* null mice by increasing persistent DNA damage and oxidative stress. *Journal of Cellular and Molecular Medicine*, 23(12).

[16] Hurt, J. A., Robertson, A. D., and Burge, C. B. (2013). Global analyses of UPF1 binding and function reveal expanded scope of nonsense-mediated mRNA decay. *Genome Research*, 23(10):1636–1650.

[17] Huth, M., Santini, L., Galimberti, E., Ramesmayer, J., Titz-Teixeira, F., Sehlke, R., Oberhuemer, M., Stummer, S., Herzog, V., Garmhausen, M., Romeike, M., Chugunova, A., Leesch, F., Holcik, L., Weipoltshammer, K., Lackner, A., Schoefer, C., Haeseler, A. v., Buecker, C., Pauli, A., Ameres, S. L., Smith, A., Beyer, A., and Leeb, M. (2022). NMD is required for timely cell fate transitions by fine-tuning gene expression and regulating translation. *Genes & Development*, 36(5-6):348–367. Company: Cold Spring Harbor Laboratory Press Distributor: Cold Spring Harbor Laboratory Press Institution: Cold Spring Harbor Laboratory Press Label: Cold Spring Harbor Laboratory Press Publisher: Cold Spring Harbor Lab.

[18] Isken, O. and Maquat, L. E. (2007). Quality control of eukaryotic mRNA: safeguarding cells from abnormal mRNA function. *Genes & Development*, 21(15):1833–1856.

[19] Jeyarajah, M. J., Jaju Bhattad, G., Kelly, R. D., Baines, K. J., Jaremek, A., Yang, F.-H. P., Okae, H., Arima, T., Dumeaux, V., and Renaud, S. J. (2022). The multifaceted role of gcm1 during trophoblast differentiation in the human placenta. *Proceedings of the National Academy of Sciences*, 119(49).

[20] Johnson, J. L., Stoica, L., Liu, Y., Zhu, P. J., Bhattacharya, A., Buffington, S. A., Huq, R., Eissa, N. T., Larsson, O., Porse, B. T., and et al. (2019). Inhibition of upf2-dependent nonsense-mediated decay leads to behavioral and neurophysiological abnormalities by activating the immune response. *Neuron*, 104(4).

[21] Jolly, L. A., Homan, C. C., Jacob, R., Barry, S., and Gecz, J. (2013). The upf3b gene, implicated in intellectual disability, autism, adhd and childhood onset schizophrenia regulates neural progenitor cell behaviour and neuronal outgrowth. *Human Molecular Genetics*, 22(23):4673–4687.

[22] Kashima, I., Yamashita, A., Izumi, N., Kataoka, N., Morishita, R., Hoshino, S., Ohno, M., Dreyfuss, G., and Ohno, S. (2006). Binding of a novel SMG-1–Upf1–eRF1–eRF3 complex (SURF) to the exon junction complex triggers Upf1 phosphorylation and nonsense-mediated mRNA decay. *Genes & Development*, 20(3):355–367. Company: Cold Spring Harbor Laboratory Press Distributor: Cold Spring Harbor Laboratory Press Institution: Cold Spring Harbor Laboratory Press Label: Cold Spring Harbor Laboratory Press Publisher: Cold Spring Harbor Lab.

[23] Keller, G. M. (1995). In vitro differentiation of embryonic stem cells. *Current Opinion in Cell Biology*, 7(6):862–869.

[24] Kubaczka, C., Kaiser, F., and Schorle, H. (2017). Breaking the first lineage barrier – many roads to trophoblast stem cell fate. *Placenta*, 60:S52–S56.

[25] Kunath, T., Strumpf, D., and Rossant, J. (2004). Early trophoblast determination and stem cell maintenance in the mouse—a review. *Placenta*, 25:S32–S38.

[26] Kurosaki, T., Popp, M. W., and Maquat, L. E. (2019). Quality and quantity control of gene expression by nonsense-mediated mRNA decay. *Nature Reviews Molecular Cell Biology*, 20(7):406–420. Number: 7 Publisher: Nature Publishing Group.

[27] Lackner, A., Sehlke, R., Garmhausen, M., Giuseppe Stirparo, G., Huth, M., Titz-Teixeira, F., Lelij, P., Ramesmayer, J., Thomas, H. F., Ralser, M., Santini, L., Galimberti, E., Sarov, M., Stewart, A. F., Smith, A., Beyer, A., and Leeb, M. (2021). Cooperative genetic networks drive embryonic stem cell transition from naïve to formative pluripotency. *The EMBO Journal*, 40(8).

[28] Lai, H.-C., James, A., Luff, J., De Souza, P., Quek, H., Ho, U., Lavin, M. F., and Roberts, T. L. (2020). Regulation of rna degradation pathways during the lipopolysaccharide response in macrophages. *Journal of Leukocyte Biology*, 109(3):593–603.

[29] Laumonnier, F., Nguyen, L. S., Jolly, L., Raynaud, M., and Gecz, J. (2014). Upf3b gene and nonsense-mediated mrna decay in autism spectrum disorders. *Comprehensive Guide to Autism*, page 1663–1678.

[30] Li, H., Zhao, P., Xu, Q., Shan, S., Hu, C., Qiu, Z., and Xu, X. (2016). The autism-related gene snrpn regulates cortical and spine development via controlling nuclear receptor nr4a1. *Scientific Reports*, 6(1).

[31] Li, T., Shi, Y., Wang, P., Guachalla, L. M., Sun, B., Joerss, T., Chen, Y., Groth, M., Krueger, A., Platzer, M., and et al. (2015). Smg6/est1 licenses embryonic stem cell differentiation via nonsense-mediatedmrnadecay. *The EMBO Journal*, 34(12):1630–1647.

[32] Li, Y., Ge, X., Peng, F., Li, W., and Li, J. J. (2022). Exaggerated false positives by popular differential expression methods when analyzing human population samples. *Genome Biology*, 23(1).

[33] Lloyd, J., Lang, D., Zimmer, A. D., Causier, B., Reski, R., and Davies, B. (2018). The loss of SMG1 causes defects in quality control pathways in Physcomitrella patens. *Nucleic Acids Research*, 46(11):5822–5836.

[34] Lloyd, J. P. B. and Davies, B. (2013). Smg1 is an ancient nonsense-mediated mrna decay effector. *The Plant Journal*, 76(5):800–810.

[35] Lou, C.-H., Chousal, J., Goetz, A., Shum, E. Y., Brafman, D., Liao, X., Mora-Castilla, S., Ramaiah, M., Cook-Andersen, H., Laurent, L., and et al. (2016). Nonsense-mediated rna decay influences human embryonic stem cell fate. *Stem Cell Reports*, 6(6):844–857.

[36] Martello, G. and Smith, A. (2014). The nature of embryonic stem cells. *Annual Review of Cell and Developmental Biology*, 30:647–675.

[37] McGlincy, N. J. and Ingolia, N. T. (2017). Transcriptome-wide measurement of translation by ribosome profiling. *Methods (San Diego, Calif.)*, 126:112–129.

[38] McIlwain, D. R., Pan, Q., Reilly, P. T., Elia, A. J., McCracken, S., Wakeham, A. C., Itie-Youten, A., Blencowe, B. J., and Mak, T. W. (2010). Smg1 is required for embryogenesis and regulates diverse genes via alternative splicing coupled to nonsense-mediated mRNA decay. *Proceedings of the National Academy of Sciences of the United States of America*, 107(27):12186–12191.

[39] Medghalchi, S. M. (2001). Rent1, a trans-effector of nonsense-mediated mrna decay, is essential for mammalian embryonic viability. *Human Molecular Genetics*, 10(2):99–105.

[40] Metze, S., Herzog, V. A., Ruepp, M.-D., and Mühlemann, O. (2013). Comparison of EJC-enhanced and EJC-independent NMD in human cells reveals two partially redundant degradation pathways. *RNA*, 19(10):1432–1448.

[41] Nguyen, L. S., Jolly, L., Shoubridge, C., Chan, W. K., Huang, L., Laumonnier, F., Raynaud, M., Hackett, A., Field, M., Rodriguez, J., and et al. (2011). Transcriptome profiling of upf3b/nmd-deficient lymphoblastoid cells from patients with various forms of intellectual disability. *Molecular Psychiatry*, 17(11):1103–1115.

[42] Nichols, J. and Jones, K. (2017). Derivation of mouse embryonic stem (es) cell lines using small-molecule inhibitors of erk and gsk3 signaling (2i). *Cold Spring Harbor Protocols*, 2017(5).

[43] Nickless, A., Bailis, J. M., and You, Z. (2017). Control of gene expression through the nonsense-mediated rna decay pathway. *Cell Bioscience*, 7(1).

[44] Nowotschin, S., Setty, M., Kuo, Y.-Y., Liu, V., Garg, V., Sharma, R., Simon, C. S., Saiz, N., Gardner, R., Boutet, S. C., Church, D. M., Hoodless, P. A., Hadjantonakis, A.-K., and Pe'er, D. (2019). The emergent landscape of the mouse gut endoderm at single-cell resolution. *Nature*, 569(7756):361–367. Number: 7756 Publisher: Nature Publishing Group.

[45] Perez-Garcia, V., Fineberg, E., Wilson, R., Murray, A., Mazzeo, C. I., Tudor, C., Sienerth, A., White, J. K., Tuck, E., Ryder, E. J., Gleeson, D., Siragher, E., Wardle-Jones, H., Staudt, N., Wali, N., Collins, J., Geyer, S., Busch-Nentwich, E. M., Galli, A., Smith,

J. C., Robertson, E., Adams, D. J., Weninger, W. J., Mohun, T., and Hemberger, M. (2018). Placentation defects are highly prevalent in embryonic lethal mouse mutants. *Nature*, 555(7697):463–468. Number: 7697 Publisher: Nature Publishing Group.

[46] Roberts, T. L., Ho, U., Luff, J., Lee, C. S., Apte, S. H., MacDonald, K. P. A., Raggat, L. J., Pettit, A. R., Morrow, C. A., Waters, M. J., Chen, P., Woods, R. G., Thomas, G. P., St. Pierre, L., Farah, C. S., Clarke, R. A., Brown, J. A. L., and Lavin, M. F. (2013). Smg1 haploinsufficiency predisposes to tumor formation and inflammation. *Proceedings of the National Academy of Sciences*, 110(4):E285–E294. Publisher: Proceedings of the National Academy of Sciences.

[47] Tanaka, S., Kunath, T., Hadjantonakis, A. K., Nagy, A., and Rossant, J. (1998). Promotion of trophoblast stem cell proliferation by FGF4. *Science (New York, N.Y.)*, 282(5396):2072–2075.

[48] Thomson, J. A., Itskovitz-Eldor, J., Shapiro, S. S., Waknitz, M. A., Swiergiel, J. J., Marshall, V. S., and Jones, J. M. (1998). Embryonic stem cell lines derived from human blastocysts. *Science*, 282(5391):1145–1147.

[49] Usuki, F., Yamashita, A., Kashima, I., Higuchi, I., Osame, M., and Ohno, S. (2006). Specific inhibition of nonsense-mediated mrna decay components, smg-1 or upf1, rescues the phenotype of ullrich disease fibroblasts. *Molecular Therapy*, 14(3):351–360.

[50] Weischenfeldt, J., Damgaard, I., Bryder, D., Theilgaard-Mönch, K., Thoren, L. A., Nielsen, F. C., Jacobsen, S. E. W., Nerlov, C., and Porse, B. T. (2008). NMD is essential for hematopoietic stem and progenitor cells and for eliminating by-products of programmed DNA rearrangements. *Genes & Development*, 22(10):1381–1396.

[51] Weischenfeldt, J., Lykke-Andersen, J., and Porse, B. (2005). Messenger RNA surveillance: neutralizing natural nonsense. *Current biology: CB*, 15(14):R559–562.

[52] Wilczynska, A., Gillen, S. L., Schmidt, T., Meijer, H. A., Jukes-Jones, R., Langlais, C., Kopra, K., Lu, W.-T., Godfrey, J. D., Hawley, B. R., and et al. (2019). Eif4a2 drives repression of translation at initiation by ccr4-not through purine-rich motifs in the 5utr. *Genome Biology*, 20(1).

[53] Wittkopp, N., Huntzinger, E., Weiler, C., Saulière, J., Schmidt, S., Sonawane, M., and Izaurralde, E. (2009). Nonsense-mediated mrna decay effectors are essential for zebrafish embryonic development and survival. *Molecular and Cellular Biology*, 29(13):3517–3528.

[54] Yamashita, A. (2013). Role of SMG-1-mediated Upf1 phosphorylation in mammalian nonsense-mediated mRNA decay. *Genes to Cells: Devoted to Molecular & Cellular Mechanisms*, 18(3):161–175.

[55] Yamashita, A., Ohnishi, T., Kashima, I., Taya, Y., and Ohno, S. (2001). Human smg-1, a novel phosphatidylinositol 3-kinase-related protein kinase, associates with components of the mrna surveillance complex and is involved in the regulation of nonsense-mediated mrna decay. *Genes Development*, 15(17):2215–2228.

[56] Zhao, X., Nogawa, A., Matsunaga, T., Takegami, T., Nakagawa, H., and Ishigaki, Y. (2014). Proteasome inhibitors and knockdown of SMG1 cause accumulation of Upf1 and Upf2 in human cells. *International Journal of Oncology*, 44(1):222–228.

[heading=bibintoc, title=References]

Appendix A

[Genes investigated]

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
Smg7	Gnai3	Zyx	Aspn
Upf2	Sdhd	Sfrs10	Mmp9
Smg5	AK041755	Alkbh3	Mgp
Smg9	Rdx	Hnrnpa2b1	Ddx3y
Smg1	Hk2	Tmem183a	Tnmd
Upf3a	Polr3d	Flot1	Myl1
Upf1	Sf3a1	Nktr	Tnnc2
Smg6	Ykt6	Mgea5	Myh3
Smg8	Wbscr9	Gtf2f2	Mybpc1
Etf1	Ppp5c	Srrm1	Tmem8c
Gstp1	Kpna1	Gls	Myog
Upf3b	Mrpl49	U2af1	Ptprb
	Ube2g2	Hnrnph1	Mylpf
	Rassf1	Rnf149	Sgcd
	Igf2bp1	Nbr1	LOC676912
	Rnf168	Sf3b1	Tnni1
	Dnajc7	Trim24	Actc1
	Csf1	1810055E12Rik	LOC100048431
	SASH1	Tatdn1	Chrna1
	Kctd5	Sf3b3	LOC100046566
	Sdc4	Snx13	Mybph
	Ywhab	Wdfy2	Cxcl12
	Akap1	Rbm5	Myl4

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
	Snw1	Chpf	Chrng
	Esrrb	Pdia5	Vnn1
	Eaf1	Abhd5	Slc1a6
	Ccdc25	Dnajc5	Synpo2l
	Xpo7	Sec61a2	Actn2
	Dnajc3	Slc37a3	Fgf7
	Parvb	Cnot2	Six2
	Slc38a2	Bclaf1	Mfap4
	Rrn3	Ubqln1	Mab21l2
	Fyttd1	Ppp4r1	Tnnc1
	Gsk3b	Usp16	Cxcl12
	Smarcd1	Nup35	Cxcl12
	Mllt1	Hspa13	Myoz2
	Stard4	Trp53bp1	Hrc
	Ddb1	Rexo4	1110059M19Rik
	Aldh18a1	l7Rn6	Capn6
	Pcgf6	Spg7	Tnnt2
	Scd2	AA536717	Bgn
	Tsr2	2610204M08Rik	Col3a1
	Gnl3l	Maz	Ltbp2
	Pa2g4	Golph3l	Svep1
	Ttc4	Tgds	Tceal7
	Crcp	Exosc10	Dnm3os
	Gm10012	Sat1	Fap
	Hgs	Eif4a2	Igf2as
	Rassf3	Rnf14	Shisa2
	Atic	Snrpa1	Thbd
	Itm2c	Rnf114	Mef2c
	nek2	Ptbp1	Acta1
	Gtf3c5	Hnrpll	Eng
	Gpd2	Eif4a2	Srl
	Lin7c	Acaa1a	Col1a2
	Wdr76	Zfp410	Igf2
	Slc23a2	Lgtn	Prss35

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
	Ldlr	Riok1	Tmem119
	Fam195b	Crls1	Myh11
	Fbxo21	Bptf	Igf1
	Pex5	Jmjd1a	Igf1
	Eif4h	Ube2h	Foxc1
	Clic4	Gria3	Igf1
	A230046K03Rik	Zc3h14	Igf1
	Exosc2	4933407N01Rik	Igf1
	Thbs1	Mtx1	Igf1
	mKIAA0376	Nfyb	Igf1
	Ccdc21	Rps9	Igf1
	Ankrd54	Golga4	Cdo1
	Scd1	Wbscr22	9130213B05Rik
	Serpine1	Topbp1	Actg2
	Aagab	Gak	Matn4
	Pdxk	Pum1	Ldb3
	Usp22	Rnps1	Ldb3
	Asb6	Bclaf1	Ldb3
	Lmtk2	Brd2	Ldb3
	Gadd45a	Hexa	Ldb3
	Pvr	Slc6a6	Pdgfrb
	Smg7	Mgrn1	Jph2
	F11r	Usp7	Itga11
	Lasp1	Becn1	Loxl4
	Tbc1d1	Klf6	Rab3il1
	Actr1b	Mtmr10	Cspg4
	AK158379	Ubp1	Igfbp6
	Eif2c2	Wsb1	Plac8
	Tbc1d13	Zmiz2	Eno3
	Wdr18	Sgce	Fam174b
	Smg6	Pcolce	Myo1d
	4F2 lc	Chordc1	Dlk2
	Ddx21	2610101N10Rik	Kcnab1
	Slc16a1	2310037I24Rik	Nnmt

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
	S1pr2	Pkp4	1500005K14Rik
	Ptpn11	Rbm5	Scn3b
	Gpr107	Ibtk	Adamts2
	Zfp518b	Kcmf1	Cpz
	Acap2	Ercc3	Lgals9
	Dcaf7	Usp32	LOC100047173
	Pitpnb	Clk1	Flt1
	Ubqln1	Ccdc100	Ppp1r14a
	Edem3	C030004A17Rik	LOC100044482
	Pygo2	Hnrnpd	Tub
	Rabgap1	Sfrs7	Kif15
	1810026J23Rik	Ipo9	LOC667382
	Zfp217	Slc25a24	Slc9a7
	Mtap1b	Cops4	Wscd1
	Nudcd3	Pik3c2a	LOC100048362
	Vapb	Zmiz1	Rasgef1b
	Trib1	Tbc1d9b	Sirt3
	Rara	Eya3	Bdkrb2
	Rlim	Clpb	LOC100046333
	Nf1	Strn3	Dhfr
	Commd7	Rcbtb2	LOC100046120
	Myst2	Rsl1d1	Ankle1
	Atxn7l3	Nbr1	Serpinb1a
	Fnip2	Sfrs2ip	Zfp423
	Pmpca	Xpnpep1	Rimklb
	Gpn1	Herc4	1810032O08Rik
	Tcfe3	Snrpb	Spc24
	Unc5b	Numa1	Plek
	Rpl7l1	Rnps1	Megf11
	Tor3a	Pkn1	LOC630164
	Sult4a1	Lancl1	Aard
	Pgd	Jmjd1b	Cxadr
	Rqcd1	Vps24	AI646023
	Ncoa3	Adss	Slc22a23

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
	B3gnt7	Nt5c3	Amigo1
	Tmbim1	Lyar	2900011O08Rik
	Wdr5	Farsb	Pde1b
	Diap1	Oxsr1	LOC100043986
	Rnf44	Cpsf6	Trpc2
	G3bp2	E430025E21Rik	Zfp536
	mKIAA1835	Ulk1	Bmp6
	Erc1	Tcf12	6330407J23Rik
	n/a	Mrpl13	Thsd7a
	Tle3	2310021P13Rik	Crb2
	Zfp445	Cflar	Zic2
		Rnf10	Rtn1
		Mtdh	Rab11fip4
		Hsf1	Chodl
		Ddx42	5530601H04Rik
		March$_5$	BC005561
		Yipf3	C630004H02Rik
		Pik3c2a	EG666856
		Hnrnpk	Fgfr3
		Tmem55a	F5
		Pigs	Samd14
		BC027231	Gabre
		1300010M03Rik	2410006H16Rik
		Wdr26	Shh
		Trappc6b	LOC668443
		Smc6	Zic1
		Snx4	LOC100048143
		Pmm2	LOC100044570
		Cog4	Hck
		Tial1	LOC100041797
		Tfdp1	LOC100045687
		Stra13	Sprr1a
		Cd47	Cyp39a1
		App	Nrxn1

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Stt3a	Slc1a3
		Ilf3	Adamts15
		Rbm39	Hmgn2
		Uvrag	Neto2
		Zbtb8os	EG665536
		Polb	Hook1
		Ilf3	Wnk2
		Azi2	Gadd45g
		Pcmt1	LOC627375
		Znrf1	EG666103
		Topbp1	LOC100047827
		Cops7b	Cxcr4
		Bace1	Mapk4
		Scrib	Rnf182
		Wdr82	LOC100046972
		Sae1	Fgfbp1
		Mkln1	Rassf4
		Ube3b	LOC100048537
		Ptk2	LOC637089
		Epha4	Heph
		Uqcrc2	Plch1
		Hnrnpl	Cnr1
		Ext2	Slc5a7
		Tank	Mapk10
		Supt6h	B230206F22Rik
		Sidt2	Msi1
		Pdcd2l	AI593442
		Nol9	4632415L05Rik
		Mrpl48	Itgb8
		Nbr1	Hap1
		Exoc5	Sox8
		Eif4g3	Epas1
		Cdk5	Ap1s3
		Brap	1810032O08Rik

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Pml	Mt3
		Mnat1	LOC100048816
		Mtch1	Podxl
		2510006D16Rik	Ascl1
		Rabep1	4930422G04Rik
		Afg3l1	Car3
		Cox17	Gdpd2
		Tmem57	Gm12824
		Rad50	Smpd3
		Mphosph8	LOC230628
		Lipa	Sema5b
		D6Wsu176e	Ppp1r1a
		Cmas	Tmem56
		Rab18	Prelp
		Mcoln1	Gdf15
		Eprs	Gpr64
		AI314180	LOC100044227
		Arfrp1	2610109H07Rik
		Ehmt2	Gna14
		Nucb2	Nsl1
		Fubp3	Folr1
		Ict1	LOC100044686
		Atxn2	Bex4
		Vps41	Zbtb8b
		Ryk	Cacnb4
		Acbd5	1500012F01Rik
		Snx6	Fam167a
		Sfrs9	LOC100043261
		Vps24	Scg2
		Pxdn	Zic4
		Prkrir	Pcp4l1
		Wipi2	Sox2
		Pomgnt1	Prokr1
		Cdc16	2700099C18Rik

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Mobkl3	Ifitm1
		Irak1	Otx2
		Cops5	Cp
		Ube2w	Slco1a5
		Spag9	St8sia1
		Rab28	Arsi
		Polr2g	LOC677369
		Pik3c2a	Prkcq
		P4ha1	A2m
		Csnk1g3	LOC100047091
		Becn1	Tmem163
		Atp6v1c1	Zic3
		Aftph	Scn5a
		2310047O13Rik	LOC100044302
		Prpf4b	Erdr1
		Wdr35	Rgs8
		Tfdp1	Iapp
		Phca	Sprr2b
		Gbf1	
		Fez2	
		Atp9b	
		Arl2bp	
		1500035H01Rik	
		Fgfr1	
		Bnip2	
		Ap1m1	
		Napa	
		Mosc2	
		Cct8	
		Tmem131	
		Tubg2	
		Snx2	
		Smarca2	
		6430706D22Rik	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Acot9	
		Prdx3	
		Ppp2r3c	
		Mtch2	
		Dusp11	
		Cyb5r4	
		Cse1l	
		Atp9b	
		Ttyh3	
		Copz2	
		Adam9	
		Nt5c3	
		Txndc4	
		Fkbp1a	
		Yars	
		Ergic1	
		Cmpk1	
		Stard7	
		Sptlc2	
		Papola	
		Mybbp1a	
		Ubxn2a	
		Tsg101	
		Slc25a17	
		Ppil2	
		Tkt	
		Tcf20	
		Srrm1	
		Rap1a	
		Cox19	
		Ankhd1	
		Agrn	
		Trim27	
		Ube2g1	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Ergic3	
		Elp2	
		Cttn	
		Ciapin1	
		AI314976	
		Dicer1	
		Immt	
		Strn3	
		Pdrg1	
		Idh3a	
		1300010M03Rik	
		Xpo6	
		Stx16	
		P4ha1	
		Ik	
		Copg	
		Ankrd13c	
		Zc3h14	
		Phip	
		Nol5	
		Med21	
		Hnrnpm	
		Ehmt2	
		Atp6v0a1	
		Crlf1	
		Txndc10	
		Tmem123	
		Puf60	
		2900010J23Rik	
		Pcgf3	
		Stam	
		Pcm1	
		Dis3	
		Cenpt	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Surf4	
		Ndufa7	
		Srp72	
		Pcnx	
		Luc7l2	
		Slc38a10	
		2810428I15Rik	
		Srp19	
		Elf2	
		Papola	
		Ilf3	
		Galnt1	
		Ddr1	
		Cars	
		Bat3	
		2610024E20Rik	
		Wdr42a	
		Galnt1	
		Bax	
		Zmym2	
		Supt5h	
		Psmc6	
		Ipo13	
		Nexn	
		Chfr	
		Epb4.1l3	
		Ndrg1	
		Mett10d	
		Glrx3	
		Dkc1	
		Tcf12	
		Slc38a10	
		Rela	
		Cnot1	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Ufd1l	
		Rrm1	
		Pepd	
		Efha1	
		Skiv2l2	
		Tram1	
		Eif5	
		Wac	
		Stam	
		Slc38a10	
		Tpd52	
		Rpn2	
		Atp6ap1	
		Npepps	
		Stk38	
		Gosr2	
		Fn1	
		Fkbp1a	
		Camk2g	
		Ivns1abp	
		Rnaset2a	
		Ppp5c	
		Ppil2	
		Pa2g4	
		D6Wsu176e	
		Canx	
		Rangap1	
		Kras	
		Rsu1	
		Anxa3	
		Yaf2	
		Tmem16f	
		Cpsf3l	
		Hmbs	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Tsg101	
		Nars	
		Laptm4a	
		Dpf2	
		Cetn3	
		Wnk1	
		Sepw1	
		Hs2st1	
		Ciapin1	
		Sypl	
		Spop	
		Dlg1	
		Vps35	
		Col11a1	
		Synj2bp	
		1810020G14Rik	
		Lsm14b	
		Gpx4	
		Hras1	
		Ssr4	
		Odz3	
		Ubap2	
		l7Rn6	
		Hsd17b12	
		Sars	
		Rbm26	
		Nptn	
		Gtf3c1	
		Bag1	
		Rock2	
		Atp5j2	
		Parva	
		Rars	
		Papss1	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Wdr45l	
		Wdfy1	
		Vta1	
		Vps8	
		Vps54	
		Utp18	
		Utp11l	
		Usp4	
		Usp39	
		Usp33	
		Unc84a	
		Unc45a	
		Ugcgl1	
		Ufsp2	
		Ufd1l	
		Ubxn4	
		Ube4a	
		Ube2k	
		Ube2k	
		Ube2g2	
		Ube2b	
		Ubap1	
		Uba3	
		Uba2	
		Uba2	
		Uap1	
		Txndc9	
		Txndc4	
		Ttc3	
		Tsr1	
		Tsr1	
		Trim33	
		Tpd52l2	
		Tomm70a	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Tapt1	
		Suz12	
		Supt5h	
		Stx8	
		Stx7	
		Stk17b	
		Spred2	
		Sppl3	
		Spg7	
		Spg7	
		Snx6	
		Snx19	
		Snd1	
		Snapin	
		Smchd1	
		Smc1a	
		Smarcad1	
		Slmo2	
		Slc4a1ap	
		Slc3a2	
		Slc38a2	
		Slc35b1	
		Slc30a5	
		Slc25a39	
		Slc25a3	
		Slc25a13	
		Shcbp1	
		Sh3glb2	
		Sfrs5	
		Sfrs2ip	
		Setd2	
		Senp6	
		Selm	
		Sec24c	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Rnf214	
		Rnaset2a	
		Rnasen	
		Rhobtb3	
		Rdh10	
		Rcn2	
		Rbm5	
		Rbm38	
		Rbm25	
		Rbbp4	
		Ranbp9	
		Ranbp3	
		Rad50	
		Rac1	
		Rabep1	
		Rab2a	
		Qdpr	
		Qars	
		Ptprs	
		Ptprf	
		Ptprf	
		Ptpn14	
		Pspc1	
		Psmd11	
		Psmd11	
		Psmc2	
		Psat1	
		Psap	
		Psap	
		Prmt7	
		Prmt3	
		Prmt2	
		Prmt1	
		Prdx5	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Pip5k1a	
		Pigp	
		Pigk	
		Pigk	
		Picalm	
		Pgs1	
		Pgm3	
		Pfkl	
		Pfdn5	
		Pex3	
		Pex19	
		Pcyox1	
		Pcnxl3	
		Parp8	
		Papss1	
		Papola	
		Pank3	
		Pak1ip1	
		Pafah1b1	
		Osbpl3	
		Orc6l	
		Nxf1	
		Nup153	
		Nup107	
		Nup107	
		Nudt5	
		Nsmaf	
		Nrd1	
		Nptxr	
		Nptn	
		Npr2	
		Npepps	
		Npepps	
		Nol5	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Mrps15	
		Mocs2	
		Mllt10	
		Mia3	
		Mgrn1	
		Mfsd11	
		Mff	
		Mff	
		Metap2	
		Med20	
		Me2	
		Mdm4	
		Mast2	
		Map2k1ip1	
		Map2k1	
		Map1lc3a	
		Man2b1	
		Man1a2	
		Mak10	
		Magt1	
		Magohb	
		Lztr1	
		Lpcat1	
		Lims1	
		Lass2	
		Laptm4a	
		Kif2a	
		Kctd20	
		Kctd10	
		Kank2	
		Jup	
		Jmjd6	
		Jmjd2a	
		Jak1	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Hdgfrp2	
		Hdac3	
		Hdac3	
		Hccs	
		Hbs1l	
		Hax1	
		H2-K1	
		H13	
		H13	
		Gtse1	
		Gtf3c2	
		Gtf3c1	
		Gtf2f1	
		Gstp2	
		Gspt1	
		Gpn3	
		Gosr1	
		Gnl3	
		Gmfb	
		Glod4	
		Ggcx	
		Gdi2	
		Ganab	
		Galnt10	
		Gak	
		Gak	
		Fus	
		Fubp3	
		Fubp3	
		Flii	
		Fkbp2	
		Fgfr2	
		Fgfr1op2	
		Fbxo3	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Emg1	
		Elmo2	
		Elavl1	
		Eif5	
		Eif3h	
		Eif3b	
		Eif2b1	
		Ehmt2	
		Efr3a	
		Dst	
		Drg1	
		Dpp8	
		Dnajc7	
		Dnajc2	
		Dnajc11	
		Dlst	
		Dlgap4	
		Dhx30	
		Dgka	
		Dgka	
		Decr2	
		Ddx17	
		D5Wsu178e	
		D15Ertd621e	
		D0HXS9928E	
		D030074E01Rik	
		Cyfip1	
		Cwc15	
		Ctps2	
		Ctps2	
		Ctbp1	
		Ctage5	
		Cstf2	
		Csnk1d	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Clu	
		Clstn1	
		Clk3	
		Chfr	
		Cdc20	
		Cd9	
		Cct7	
		Ccnc	
		Cbfb	
		Cand1	
		Cacnb3	
		C80913	
		C330027C09Rik	
		C330007P06Rik	
		C030004A17Rik	
		Bsg	
		Brd7	
		Bptf	
		Bclaf1	
		Bcl7c	
		BC016423	
		BC011248	
		Bat2d	
		Bap1	
		Azin1	
		Atrnl1	
		Atpbd4	
		Atp9b	
		Atp8b2	
		Atp6v1c1	
		Atp6ap1	
		Atic	
		Atg5	
		Ascc2	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Akap8	
		Ahsa1	
		Ahcyl1	
		Agrn	
		Agrn	
		Agpat6	
		Adsl	
		Adam19	
		Actr3	
		Actr10	
		Actl6a	
		Acox1	
		Aco2	
		Acd	
		Acadm	
		AA536717	
		A630042L21Rik	
		6330569M22Rik	
		6030443O07Rik	
		5730453I16Rik	
		4932417H02Rik	
		4932417H02Rik	
		4930402H24Rik	
		2610507B11Rik	
		2410001C21Rik	
		1300001I01Rik	
		Wdr42a	
		Vta1	
		Ubfd1	
		Tor1aip1	
		Tmem55a	
		Tmem131	
		Tmem123	
		Tle4	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Rtf1	
		Rsu1	
		Rnf115	
		Rgl1	
		Rell1	
		Reep5	
		Reep5	
		Rc3h2	
		Rbm25	
		Rangap1	
		Puf60	
		Ptpn1	
		Ptbp1	
		Psap	
		Ppp2r4	
		Ppp1r9b	
		Pigt	
		Phldb1	
		Pfkp	
		Pdlim5	
		P4ha2	
		Oxct1	
		Nt5c3	
		Ndufs4	
		Nans	
		Mtss1	
		Mtmr6	
		Mtmr3	
		Msrb3	
		Mrpl3	
		Mpp1	
		Mef2a	
		Mat2a	
		Mat2a	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Hs2st1	
		Hook3	
		Hint3	
		Gtpbp2	
		Gtf3c1	
		Gtf3c1	
		Gnpda2	
		Gdi2	
		Fndc3a	
		Fkbp14	
		Fat1	
		Erp29	
		Epb4.1	
		Eif4e	
		Dync1li2	
		Dync1i2	
		Dpf2	
		D6Wsu176e	
		Cxadr	
		Cox4i1	
		Copz1	
		Cops4	
		Cops2	
		Commd1	
		Cmpk1	
		Cltc	
		Ckap5	
		Chchd3	
		Chchd3	
		Centb2	
		Cdc16	
		Cd63	
		Bub3	
		Bre	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		1700025G04Rik	
		1600012F09Rik	
		Pigp	
		Stk25	
		Rps9	
		Fbxw5	
		Cul1	
		Supt5h	
		Sec11a	
		Ndufa12	
		Sar1a	
		Ptcd3	
		Epb4.1l3	
		Frmd4a	
		Phldb1	
		Cdc2l1	
		Uso1	
		Rab11a	
		Ddb1	
		Rexo2	
		Ccdc132	
		Axl	
		Dctn6	
		F3	
		Mest	
		Mdm4	
		Tlk1	
		2810405J04Rik	
		Nf2	
		Iqgap1	
		Sap30bp	
		Park7	
		Ing4	
		Chchd3	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Ganab	
		Gabpa	
		Brwd2	
		Gfpt1	
		Vps24	
		Twsg1	
		Ppp6c	
		Laptm4b	
		Arpc4	
		Vta1	
		Txndc1	
		Ppp1r12a	
		Pfdn5	
		Atp6ap1	
		Ptpn1	
		Arfgap1	
		Ap3b1	
		1110001A07Rik	
		Trp53	
		Ptbp1	
		Pkd1	
		Ogdh	
		Fis1	
		1110008F13Rik	
		Camk2g	
		Prmt2	
		Klf3	
		Atp2b1	
		Phf14	
		Clasp1	
		3110001A13Rik	
		Tpp1	
		Map2k4	
		Lpp	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Cdk9	
		F3	
		Zfyve27	
		Ube2g1	
		Csnk1a1	
		Sltm	
		Pmpcb	
		Nrd1	
		Ifnar2	
		Ganab	
		Sec11a	
		Snrpn	
		Rnf115	
		2610024E20Rik	
		Tia1	
		Mtfr1	
		Ttc3	
		Srp19	
		Faf2	
		Cyc1	
		Rhot1	
		Fubp3	
		Stag2	
		Traf7	
		Ss18	
		Sh3bp4	
		Commd7	
		Nfatc3	
		Rangap1	
		Lass5	
		Elp2	
		Nrd1	
		Gfpt1	
		Nubp2	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Unc5b	
		Fer1l3	
		Plk1	
		Arfgap1	
		Tial1	
		Brd8	
		Cmtm7	
		Adipor2	
		Ptprf	
		Brms1l	
		Ctbp1	
		Rhbdf1	
		Myo18a	
		Csde1	
		Npnt	
		Ptpn12	
		Mxd4	
		Ttc14	
		Snx19	
		Slc4a7	
		Eif4enif1	
		2810452K22Rik	
		Txnl4a	
		Foxk2	
		Stam2	
		Fkbp14	
		Dnajc15	
		Zfand3	
		Rbms1	
		Abhd12	
		Ttc14	
		Fubp3	
		Dpm1	
		Prmt5	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Ptprf	
		Angel2	
		Ict1	
		Ankrd17	
		Elf2	
		Usp3	
		4921524J17Rik	
		Odf2	
		Ss18	
		Ndufs4	
		Ehmt2	
		Supt5h	
		Aasdhppt	
		Rai1	
		Rps6ka2	
		Bin3	
		Slc38a10	
		Ptprf	
		Polb	
		Fchsd2	
		Stoml2	
		Bap1	
		Vps25	
		Bptf	
		Ptpla	
		Glb1	
		2510006D16Rik	
		Efemp1	
		Ube2d3	
		Sort1	
		Pkp4	
		Zc3h14	
		Rai14	
		Ktn1	

Genes in NMD pathway	UPF1 binding in 3'UTR	NMD regulated AS events	Highly DE in Smg1-/- MEFs
		Golga4	
		Myst2	
		Itga6	
		Git2	
		Magi1	
		Axl	
		Pam	
		Fn1	

Appendix B

Differentially Expressed Genes

UPF1 binding in the 3'UTR		NMD regulated AS events		Highly DE in Smg1 knockout MEFs	
TS cells	ES cells	TS cells	ES cells	TS cells	ES cells
Rassf1	Tor3a	Rdh10	Casp8	Bmp7	Col3a1
	Thbs1	Alkbh3	Kctd3	Casq2	Ap1s3
	Csf1	Ccnl1	Trp53bp1	Lrp4	Tmem163
	Dapp1	Ggcx	Wdr47	1810026B05Rik	F5
	Gpn1	Arih2	Nexn	Trpc2	Rxrg
	Gadd45a	Tatdn1	Mxd4	Sirt3	Lrp2
	Ddit4	Eif4a2	Snrpn	Scn5a	Pax6
	Rassf3		Fgfr2	2410006H16Rik	Jph2
	Slc38a2		Crlf1	2610307P16Rik	Mmp9
	Csrp1		Ifngr1	Tnnc1	Cdh4
	Tbc1d13		Specc1	Slc1a3	Car3
	Src		Pxdn	Aard	Cnr1
	Ntn1		Pdlim7		Hook1
	Cdc42ep4		Erap1		Fndc5
	Esrrb		Parp8		Fgfr3
			Nptxr		Rasgef1b
			Slc38a2		Mapk10
			Lpp		Podxl
			H2-K1		Smyd1
			Ddr1		Rassf4
			Fbn2		Cxcl12
			Pam		Tnnt1
			Rgl1		Nell1
			Frmd4a		Trim72
			Tbc1d13		H19
			Fkbp7		Sox1

UPF1 binding in the 3'UTR		NMD regulated AS events		Highly DE in Smg1 knockout MEFs	
TS cells	ES cells	TS cells	ES cells	TS cells	ES cells
			Atp8b2		Neto2
			Col11a1		Zfp423
			Agrn		Mt3
			Slit2		Zic1
			Rell1		Slc35f1
			Anxa3		Plek
			Ulk1		Sgcd
			Pcolce		Myh3
			Pdgfa		Eno3
			Fkbp14		Lgals9
			Slc6a6		Itgb8
			Cd9		2610307P16Rik
			Axl		Bmp6
			Man2a2		Gadd45g
			Tpp1		Otx2
			Tead1		Myh7
			Fbxo25		Fam167a
			Kank2		Slc1a3
			Ncam1		Mal2
			Hexa		Myh11
			Cmtm7		Sox8
			Cd63		Epas1
			Efemp1		Nrxn1
			Jup		Clcf1
			Ubxn2a		Actn3
			Lgals8		Gna14
			Serpinb9b		Sorcs1
			Ndrg1		Adrb1
			Cd47		Bgn
			Decr2		Rnf128
			Stk38		Cldn2
			Map4k3		Dcx
			Cask		Gpm6b
			Timp1		Cxcr4
			Renbp		Tnni1
			Mpp1		Bmp7
			Chm		Mme
			Sat1		Casq2
			Acot9		Gem
					Galnt12
					St8sia1
					Mylpf

UPF1 binding in the 3'UTR		NMD regulated AS events		Highly DE in Smg1 knockout MEFs	
TS cells	ES cells	TS cells	ES cells	TS cells	ES cells
					Prss35
					Mybpc1
					Efemp1
					Hap1
					Bdkrb2
					Serpinb1a
					Rnf182
					Mef2c
					Nid2
					Lrrc15
					Cyp39a1
					Cdo1
					Pdgfrb
					Capn6